Edited by
Mahar Afaq Safdar Muhammadi
MSc Math, Whatsapp 03494873115

دعاؤں میں یاد رکھیے گا۔

Department of
Mathematics & Statistics

Virtual University of Pakistan

STA 404
Regression & Correlation Analysis

By
Javed Iqbal
Department of Mathematics & Statistics
Virtual University of Pakistan

Lecture 1
Segment 1

"Knowledge not shared, is wasted." – Clan Jacobs

Welcome
to
STA 404
Regression and Correlation Analysis

Course prerequisites

Statistics & Probability

Course outline

- Simple linear regression
- Multiple linear regression
- Simple Correlation
- Multiple and Partial correlation
- Interval estimation
- Tests of hypothesis

Text/Refereces books

1. Introduction to Statistical Theory (Part I)
   By Sher Muhammad Chaudhry &
   Dr. Shahid Kamal

2. Introduction to Linear Regression Analysis,
   By Montgomery, D. C., and Peck, E.
   John Wiley & Sons

Content of Lecture 1

1-1 Basic introduction

1-2 Finding relationship through scatter plot

1-3 The concept of best fitting line

Introduction

- Regression Analysis is a statistical tool for investigation the relationship between a dependent variable & one or more independent variables.

- Regression analysis is widely used for prediction & forecasting.

Applications

- Economics
- Management
- Engineering
- Social sciences
- Medical fields
- Sports & Physical sciences
- Computer sciences
- Many others

Some background

- *Write* increase *or* decrease *to complete the sentence regarding expectations for the situation.*

1. The more one studies, their grades will_____.
2. The more a person diets, their weight will _____,
3. The more a person reads, their vocabulary will _____.
4. If winter, sales of ice cream _____.

The purpose of this Course is to answer these questions statistically:

1. Are two or more variables related?
2. If so, what is the strength of the relationship?
3. What type of relationship exists?
4. What kind of predictions can be made from the relationship?

Relationship between variables

- Exact relationship
- Fahrenheit ~ Celsius (centigrade)
- Deterministic model
- Y = a + b X

- Non-exact prelateship
- Weight ~ Height
- Probabilistic model
- Y = a + b X + e.

The END

Scatter Plot

Relationship between variables?

- If a teacher wishes to see whether there is a relationship between number of hours of study and test scores on an exam, he must select a random sample of students, determine the hours each studied, and obtain their grades on the exam.

Student	Hours of study x	Grade y (%)
A	6	82
B	2	63
C	1	57
D	5	88
E	2	68
F	3	75

Dependent Vs Independent Variable

- The two variables for this study are called the independent variable and the dependent variable.
- The independent variable is the variable that can be ***controlled or manipulated.***
- In this case, the number of hours of study is the independent variable and is denoted as the ***x*** variable.

Dependent Vs Independent Variable

- The dependent variable is the variable in regression that can NOT be controlled or manipulated.
- The grade the student received on the exam is the dependent variable, designated as the ***y*** variable.
- The reason for this distinction between the variables is that you ***assume*** that the grade the student *depends* on the number of hours the student studied.
- Also, you assume that, to some extent, the student can regulate or *control* the number of hours he or she studies for the exam.

How to choose …

- The determination of the *x* and *y* variables is not always clear-cut and is sometimes an arbitrary decision.
- For example, if a researcher studies the effects of age on a person's blood pressure, the researcher can generally assume that age affects blood pressure. Hence, the variable *age* can be called the *independent variable,* and the variable *blood pressure* can be called the *dependent variable.*
- On the other hand, if a researcher is studying the attitudes of husbands on a certain issue and the attitudes of their wives on the same issue, it is difficult to say which variable is the independent variable and which is the dependent variable. In this study, the researcher can ***arbitrarily*** designate the variables as independent and dependent.

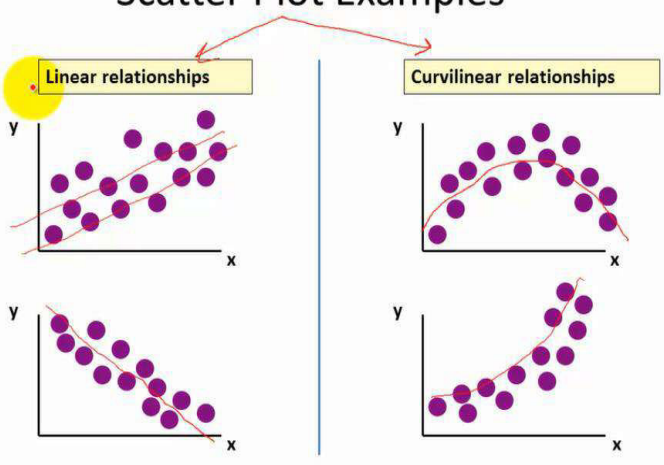
Scatter Plots and Relationship

- A scatter plot (or scatter diagram) is used to show the relationship between two variables
- It is the first step in finding the relationship between variables

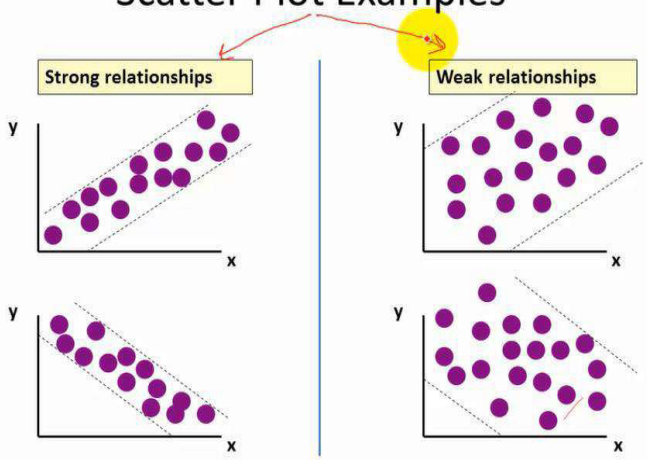
Scatter plot

- The independent and dependent variables can be plotted on a graph called a *scatter plot.*
- The independent variable *x* is plotted on the horizontal axis,
- The dependent variable *y* is plotted on the vertical axis.

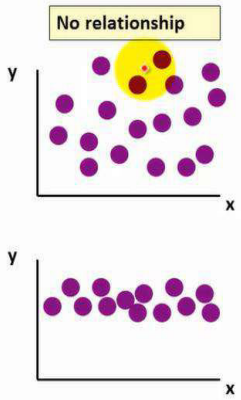
Scatter Plot Examples



Scatter Plot Examples



Scatter Plot Examples

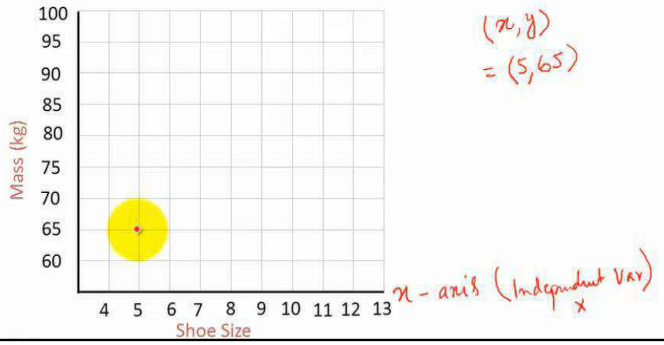


The table below shows the shoe size and weight of 8 men. Plot a scatter graph for this data.

Size (x)	5	12	7	10	10	9	8	11	6
Weight (y)	65	97	68	92	78	78	76	88	74

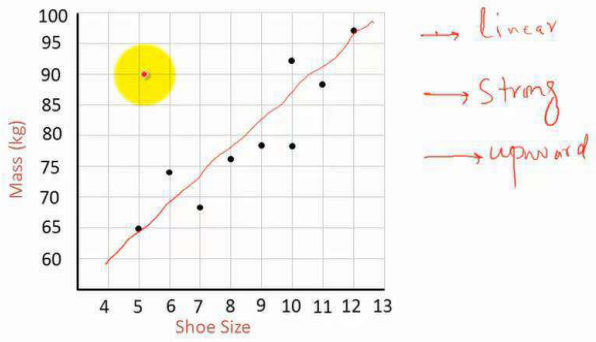
The table below shows the shoe size and weight of 8 men. Plot a scatter graph for this data.

Size (x)	5	12	7	10	10	9	8	11	6
Weight (y)	65	97	68	92	78	78	76	88	74



The table below shows the shoe size and weight of 8 men. Plot a scatter graph for this data.

Size (x)	5	12	7	10	10	9	8	11	6
Weight (y)	65	97	68	92	78	78	76	88	74



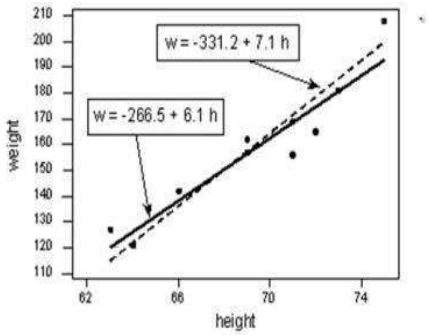
The END

What is the “best” fitting line?

What is the "Best Fitting Line"?

Student (i)	Height (x_i)	Weight (y_i)
1	63	127
2	64	121
3	66	142
4	69	157
5	69	162
6	71	156
7	71	169
8	72	731
9	73	181
10	75	08

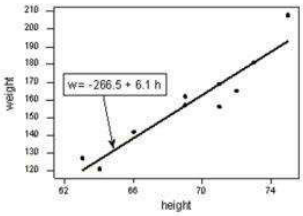
What is the "Best Fitting Line"?



Which line do you think best summarizes the trend between height and weight?

What is perdition error?

i	x_i	y_i	$\hat{y}_i$
1	63	127	120.1
2	64	121	126.3
3	66	142	138.5
4	69	157	157.0
5	69	162	157.0
6	71	156	169.2
7	71	169	169.2
8	72	165	175.4
9	73	181	181.5
10	75	208	193.8



$w = -266.5 + 6.1\,h$

-266.53 + 6.1376(63) = 120.1

It has some "**prediction error**" (or "**residual error**"). In fact, the size of its prediction error is 127-120.1 or 6.9 pounds.

Best fit?

- when we use $\hat{y}_i = a + bx_i$ to predict the actual response y_i, we make a prediction error (or residual error) of size:

$$e_i = y_i - \hat{y}_i$$

A line that fits the data "**best**" will be one for which the ***n* prediction errors — one for each observed data point — are as small as possible in some overall sense**.

One way to achieve this goal is to invoke the "**least squares criterion**," which says to "minimize the sum of the squared prediction errors."

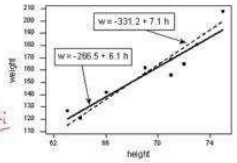
Least Square Criteria

The equation of the best fitting line is $\hat{y}_i = a + bx_i$

We just need to find the values "a" and "b" that make the sum of the squared prediction errors the smallest it can be.

That is, we need to find the values "a" and "b" that minimize: $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

- error = $e = y_i - \hat{y}_i$
- Squared of $(y_i - \hat{y}_i) \longrightarrow (y_i - \hat{y}_i)^2$
- Sum of all Sq. Error $\longrightarrow \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$



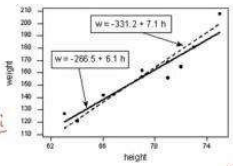
$\hat{y}_i = -266.53 + 6.1376 x_i$

w = -266.53 + 6.1376 *h* (the solid line)

i	x_i	y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)$	$(y_i - \hat{y}_i)^2$
1	63	127	120.139	6.8612	47.076
2	64	121	126.276	-5.2764	27.840
3	66	142	138.552	3.4484	11.891
4	69	157	156.964	0.0356	0.001
5	69	162	156.964	5.0356	25.357
6	71	156	169.240	-13.2396	175.287
7	71	169	169.240	-0.2396	0.057
8	72	165	175.377	-10.3772	107.686
9	73	181	181.515	-0.5148	0.265
10	75	208	193.790	14.2100	201.924
					597.4

w = -331.2 + 7.1 *h* (the dashed line)

i	x_i	y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)$	$(y_i - \hat{y}_i)^2$
1	63	127	116.1	10.9	118.81
2	64	121	123.2	-2.2	4.84
3	66	142	137.4	4.6	21.16
4	69	157	158.7	-1.7	2.89
5	69	162	158.7	3.3	10.89
6	71	156	172.9	-16.9	285.61
7	71	169	172.9	-3.9	15.21
8	72	165	180.0	-15.0	225.00
9	73	181	187.1	-6.1	37.21
10	75	208	201.3	6.7	44.89
					766.5



$\hat{y}_i = -266.53 + 6.1376 x_i$ $\quad$ $\hat{y}_i = -331.2 + 7.1 x_i$

w = -266.53 + 6.1376 *h* (the solid line)

i	x_i	y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)$	$(y_i - \hat{y}_i)^2$
1	63	127	120.139	6.8612	47.076
2	64	121	126.276	-5.2764	27.840
3	66	142	138.552	3.4484	11.891
4	69	157	156.964	0.0356	0.001
5	69	162	156.964	5.0356	25.357
6	71	156	169.240	-13.2396	175.287
7	71	169	169.240	-0.2396	0.057
8	72	165	175.377	-10.3772	107.686
9	73	181	181.515	-0.5148	0.265
10	75	208	193.790	14.2100	201.924
					597.4

w = -331.2 + 7.1 *h* (the dashed line)

i	x_i	y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)$	$(y_i - \hat{y}_i)^2$
1	63	127	116.1	10.9	118.81
2	64	121	123.2	-2.2	4.84
3	66	142	137.4	4.6	21.16
4	69	157	158.7	-1.7	2.89
5	69	162	158.7	3.3	10.89
6	71	156	172.9	-16.9	285.61
7	71	169	172.9	-3.9	15.21
8	72	165	180.0	-15.0	225.00
9	73	181	187.1	-6.1	37.21
10	75	208	201.3	6.7	44.89
					766.5

Based on the least squares criterion, which equation best summarizes the data?

Solid line vs dotted line

Can we repeat this process for every possible line?

What the solution?

The END

STA404
Regression & Correlation Analysis
Lesson 1

How to fit a Regression line?

Procedure and step by step calculation with example

How to fit a Regression line?

Example:

The following table describes data for the number of people using a swimming pool over 8 days in summer and the corresponding maximum temperature (in degrees Celsius) on each day.

Day	Temperature (^{0}C) x	Number of people y
1	20	280
2	24	360
3	36	450
4	32	420
5	28	400
6	38	500
7	32	475
8	26	320

Equation of Regression line

$$\hat{y} = a + b\,x$$

Formula for a:

$$a = \bar{y} - b(\bar{x})$$

$\bar{x} = ?$

$\bar{y} = ?$

Formula for b:

$$b = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$n = ?$

$\Sigma xy = ?$

$\Sigma x = ?$

$\Sigma y = ?$

$\Sigma x^2 = ?$

Procedure Table

Finding the Correlation Coefficient and the Regression Line Equation

Step 1 Make a table, as shown in step 2.

Step 2 Find the values of xy, x^2, and y^2. Place them in the appropriate columns and sum each column.

x	y	xy	x^2	y^2
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
$\Sigma x =$	$\Sigma y =$	$\Sigma xy =$	$\Sigma x^2 =$	$\Sigma y^2 =$

Step 3 Substitute in the formula to find the value of r.

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n(\Sigma x^2) - (\Sigma x)^2][n(\Sigma y^2) - (\Sigma y)^2]}}$$

Step 4 When r is significant, substitute in the formulas to find the values of a and b for the regression line equation $y' = a + bx$.

$$a = \bar{y} - b(\bar{x}) \qquad b = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

Required Calculation

Day	X	Y
1	20	280
2	24	360
3	36	450
4	32	420
5	28	400
6	38	500
7	34	475
8	26	320
Total	**238**	**3205**

- From previous table we have:

$n = 8$

$\Sigma x = 238$

$\Sigma y = 3205$

$\Sigma x^2 = 7356$

$\Sigma xy = 98550$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{3205}{8} = 400.62$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{238}{8} = 29.75$$

- Now putting these values in "b" and "a", and by solving, we get:

$$b = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}.$$

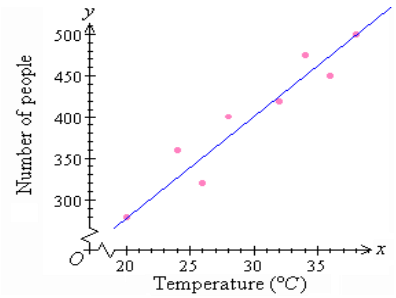
- Putting the values of "a" and "b" in the equation, we get:

$$\hat{y} = 54.92 + 11.62(x)$$

No. of People $= 54.92 + 11.62 \times$ (Temprature)

- **Interpretation:**
- This means that if the daily temperature is increased by 1 degree, we would expect the 11.62 more person to use the swimming pool.

Scatter plot



- The END -

Derivation of Least Square Estimates

Derivation of Least Square Estimates

- The line fitted by least square is the one that makes the "sum of squares of all errors" as small as possible.

$$S = SSE = \sum e_i^2 \longrightarrow \text{Minimum}$$
$$= \sum (y_i - \hat{y}_i)^2$$
$$= \sum (y_i - a - bx_i)^2 \qquad \because \hat{y}_i = a + bx_i$$

Normal Equations

To become minimum, it must fulfill the following condition.

$$S = SSE = \sum (y_i - a - bx_i)^2$$

Take partial derivative w.r.t "a" and equating to zero

$$\frac{\partial S}{\partial a} = 2\sum (y_i - a - bx_i)(0 - 1 - 0) = 0$$
$$-2\sum (y_i - a - bx_i) = 0$$
$$\sum y_i - na - b\sum x_i = 0 \Rightarrow \sum y_i = na + b\sum x_i$$

Normal Equations

To become minimum, it must fulfill the following condition.

$$S = SSE = \sum (y_i - a - bx_i)^2$$

Take partial derivative w.r.t "a" and equating to zero

$$\frac{\partial S}{\partial a} = 2\sum (y_i - a - bx_i)(0 - 1 - 0) = 0$$
$$-2\sum (y_i - a - bx_i) = 0$$
$$\sum y_i - na - b\sum x_i = 0 \Rightarrow \boxed{\sum y_i = na + b\sum x_i} \quad \text{Eq ①}$$

Similarly,

$$\frac{\partial S}{\partial b} = 2\sum (y_i - a - bx_i)(0 - 0 - x_i) = 0$$
$$-2\sum x_i (y_i - a - bx_i) = 0$$
$$\sum x_i y_i - a\sum x_i - b\sum x_i^2 = 0 \Rightarrow \boxed{\sum x_i y_i = a\sum x_i + b\sum x_i^2} \quad \text{Eq ②}$$

Note: Eq ① & Eq ② are called "Normal Equations"

Formula for "a"

From Normal Eq ①

$$\sum y_i = na + b\sum x_i$$

Dividing b.s by n

$$\frac{\sum y_i}{n} = \frac{na}{n} + \frac{b\sum x_i}{n}$$

$$\bar{y} = a + b\bar{x}$$

$$\Rightarrow \boxed{a = \bar{y} - b\bar{x}}$$

Formula for "b"

$$\sum y_i = na + b\sum x_i \quad \text{—①}$$

$$\sum x_i y_i = a\sum x_i + b\sum x_i^2 \quad \text{—②}$$

Multiply by 'n'

$$n\sum x_i y_i = na\sum x_i + nb\sum x_i^2$$

$$-\sum x_i \sum y_i = -na\sum x_i - b(\sum x_i)^2$$

$$n\sum x_i y_i - \sum x_i \sum y_i = nb\sum x_i^2 - b(\sum x_i)^2$$

$$n\sum x_i y_i - \sum x_i \sum y_i = b\{n\sum x_i^2 - (\sum x_i)^2\}$$

$$\Rightarrow \boxed{b = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}}$$

The END

STA404
Regression & Correlation Analysis
Lesson 3

What is Coefficient of Determination?

SST, SSE, SSR and CoD with example and explanation

Topic

- What is SSE?
- What is SSR?
- What is SST?
- What is Coefficient of Determination?

Example

Let's assume that you are a small restaurant owner or a very business minded server / waiter at a nice restaurant. Here in the U.S. "tips" are a very important part of a waiter's pay. Most of the time the dollar amount of tip is related to the dollar amount of the total bill.

As the waiter or owner, you would like to develop a model that will allow you to make a prediction about what amount of tip to expect. Therefore one evening, you collect data for six meals.

Tip for Service

Unfortunately when you begin to look at your data, you realize you only collected data for the tip amount and not the meal amount also! So this is the best data you have.

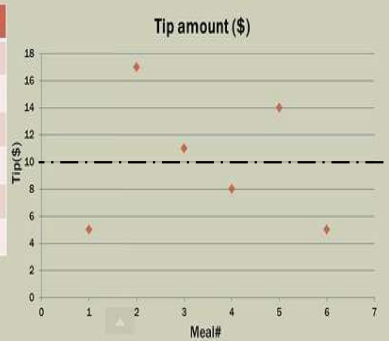
How might you predict the tip amount for future meals using only this data?

Meal #	Tip amount ($)
1	5.00
2	17.00
3	11.00
4	8.00
5	14.00
6	5.00

Tip for Service

Meal#	Tip amount ($)
1	5.00
2	17.00
3	11.00
4	8.00
5	14.00
6	5.00

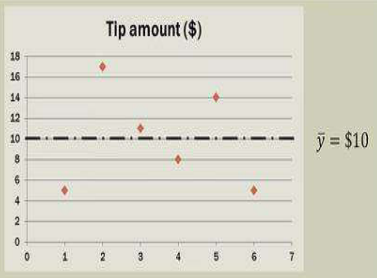
$\bar{y} = \$10$



Tip for Service

Meal#	Tip amount ($)
1	5.00
2	17.00
3	11.00
4	8.00
5	14.00
6	5.00

$\bar{y} = \$10$

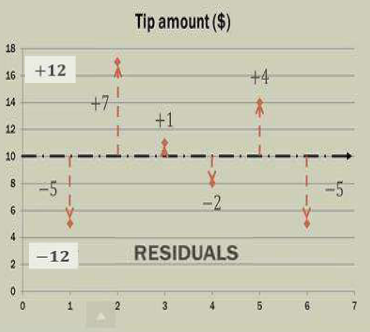


With only one variable, and no other information, the best prediction for the next measurement is the mean of the sample itself. The variability in the tip amounts can only be explained by the tips themselves.

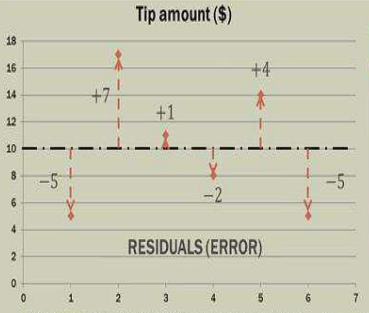
Residuals

Meal#	Tip amount ($)
1	5.00
2	17.00
3	11.00
4	8.00
5	14.00
6	5.00

$\bar{y} = \$10$



Squaring the Residuals (Errors)

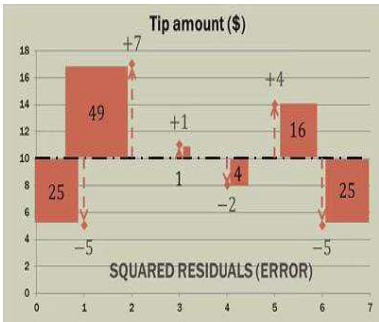


Meal#	Residual	Residual²
1	−5	25
2	+7	49
3	+1	1
4	−2	4
5	+4	16
6	−5	25

Why square the residuals? 1) makes them positive and 2) emphasizes larger deviations.

$Sum\ of\ squared\ errors\ (SSE) = 120$

Visual Look - Squares of Errors



Meal#	Residual	Residual^2
1	−5	25
2	+7	49
3	+1	1
4	−2	4
5	+4	16
6	−5	25

Sum of squared errors (SSE) = 120

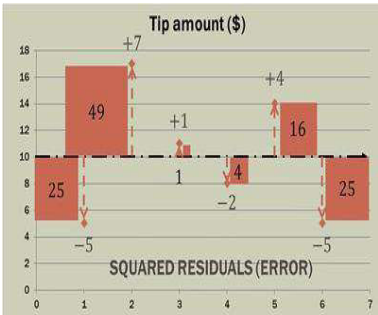
Sum of Squares of Errors

$$49 + 25 + 1 + 4 + 16 + 25 = 120$$

The goal of simple linear regression is to create a linear model that minimizes the sum of squares of the residuals / error (SSE).

If our regression model is significant, it will "eat up" much of the raw SSE we had when we assumed (like this problem) that the independent variable did not even exist. The regression line will/should literally "fit" the data better. It will minimize the residuals.

Important Note – Good Fit



Sum of squared errors (SSE) = 120

When conducting simple linear regression with TWO variables, we will determine how good that line "fits" the data by comparing it to THIS TYPE; where we pretend the second variable does not even exist.

If a two-variable regression model looks like this example, what does the other variable do to help explain the dependent variable?

NOTHING.

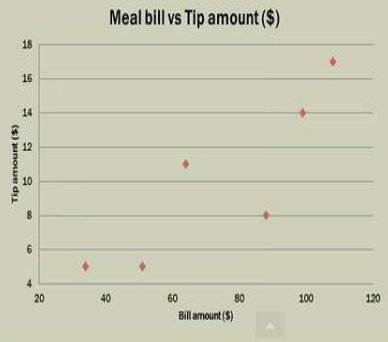
Now – Complete Data

you only had the tip data. But now you were able to go back and get the actual bill data as well. So now you are working with two variables that are matched pairs.

You want to know to what degree the tip amount can be predicted by the bill. So the tip is the DEPENDENT variable; bill is the INDEPENDENT variable.

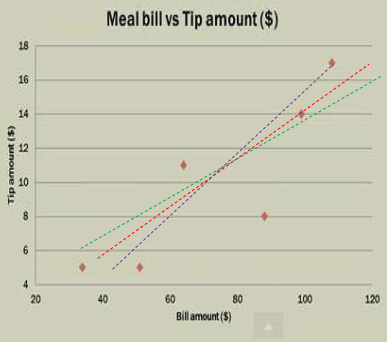
Total bill ($)	Tip amount ($)
34.00	5.00
108.00	17.00
64.00	11.00
88.00	8.00
99.00	14.00
51.00	5.00

Scatter Plot



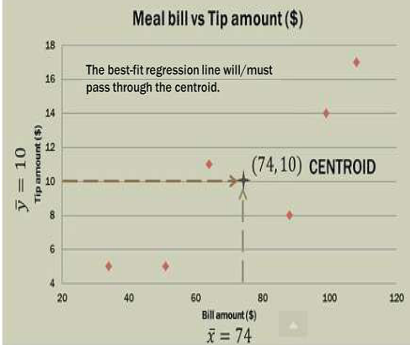
Bill ($)	Tip ($)
34.00	5.00
108.00	17.00
64.00	11.00
88.00	8.00
99.00	14.00
51.00	5.00

Look for Visual Line



Does the data seem to fall along a line?

What is the Center of the Data?



Bill ($)	Tip ($)
34.00	5.00
108.00	17.00
64.00	11.00
88.00	8.00
99.00	14.00
51.00	5.00
$\bar{x} = 74$	$\bar{y} = 10$

After Calculation

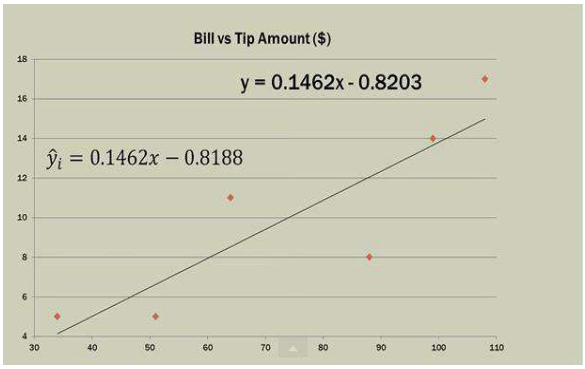
$$\hat{y}_i = a + bx$$

$$b = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n\Sigma(x^2) - (\Sigma x)^2}$$

$$a = \bar{y} - b\bar{x}$$

$$\hat{y}_i = -0.8188 + 0.1462x$$

Regression Line Drawn



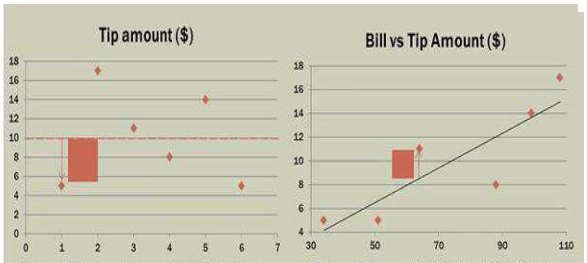
Interpretation

$$\hat{y}_i = -0.8188 + 0.1462x$$

If the bill amount (x) is zero, then the expected/predicted tip amount is $-0.8188 or negative 82-cents! Does this make sense? NO. The intercept may or may not make sense in the "real world."

For every $1 the bill amount (x) increases, we would expect the tip amount to increase by $0.1462 or about 15-cents.

Two Lines - Side by Side



$SSE = 120$

$SSE = SST$

$SST = 120$

With only the dependent variable, the only sum of squares is due to error. Therefore it is also the total, and **MAXIMUM** sum of squares for the data under analysis.

$SST = 120$

$SSE = ?$

$SST - SSE = SSR$

With both the IV and DV, the total sum of squares remains the same. But (ideally) the error sum of squares will be reduced significantly. The difference between SST and SSE is due to regression, SSR.

Estimated Regression Values

Meal	Observed Total bill ($)	Observed Tip amount ($)	$\hat{y}_i = 0.1462x - 0.8188$	$\hat{y}_i$ (predicted tip amount)
	x	y		
1	34	5		
2	108	17		
3	64	11		
4	88	8		
5	99	14		
6	51	5		
	$\bar{x} = 74$	$\bar{y} = 10$		

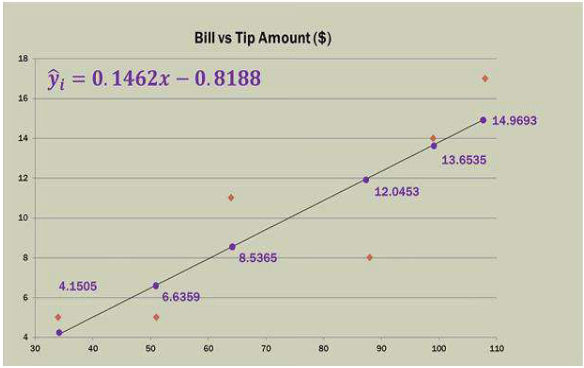
Estimated Regression Values

Meal	Total bill ($)	Tip amount ($)	$\hat{y}_i = 0.1462x - 0.8188$	$\hat{y}_i$ (predicted tip amount)
	x	y		
1	34	5	$\hat{y}_i = 0.1462(34) - 0.8188$	
2	108	17	$\hat{y}_i = 0.1462(108) - 0.8188$	
3	64	11	$\hat{y}_i = 0.1462(64) - 0.8188$	
4	88	8	$\hat{y}_i = 0.1462(88) - 0.8188$	
5	99	14	$\hat{y}_i = 0.1462(99) - 0.8188$	
6	51	5	$\hat{y}_i = 0.1462(51) - 0.8188$	
	$\bar{x} = 74$	$\bar{y} = 10$	Observed vs. Predicted	

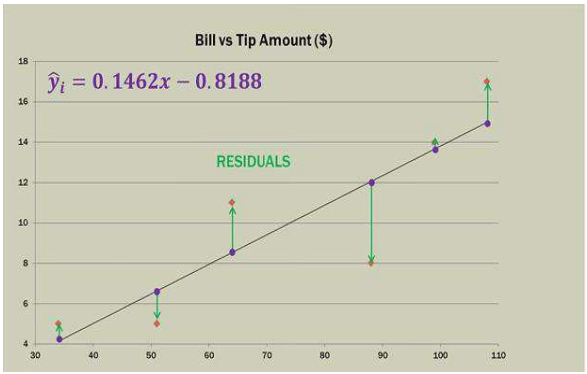
Estimated Regression Values

Meal	Total bill ($)	Tip amount ($)	$\hat{y}_i = 0.1462x - 0.8188$	$\hat{y}_i$ (predicted tip amount)
	x	y		
1	34	5	$\hat{y}_i = 0.1462(34) - 0.8188$	4.1505
2	108	17	$\hat{y}_i = 0.1462(108) - 0.8188$	14.9693
3	64	11	$\hat{y}_i = 0.1462(64) - 0.8188$	8.5365
4	88	8	$\hat{y}_i = 0.1462(88) - 0.8188$	12.0453
5	99	14	$\hat{y}_i = 0.1462(99) - 0.8188$	13.6535
6	51	5	$\hat{y}_i = 0.1462(51) - 0.8188$	6.6359
	$\bar{x} = 74$	$\bar{y} = 10$	Observed vs. Predicted	

Estimated vs Observed values



Regression Errors (Residuals)



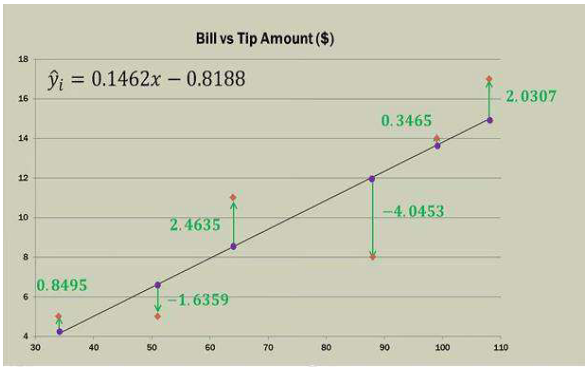
Regression Errors (Residuals)

Meal	Total bill ($)	Observed tip amount ($)	$\hat{y}_i$ (predicted tip amount)	Error: $(observed - predicted)$ $(y - \hat{y}_i)$
	x	y		
1	34	5	4.1505	$5 - 4.1505 = 0.8495$
2	108	17	14.9693	$17 - 14.9693 = 2.0307$
3	64	11	8.5365	$11 - 8.5365 = 2.4635$
4	88	8	12.0453	$8 - 12.0453 = -4.0453$
5	99	14	13.6535	$14 - 13.6535 = 0.3465$
6	51	5	6.6359	$5 - 6.6359 = -1.6359$
	$\bar{x} = 74$	$\bar{y} = 10$		

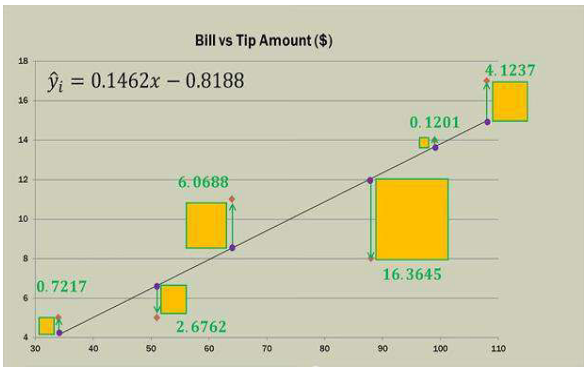
Squared Errors (Residuals)

Meal	Total bill ($)	Observed tip amount ($)	$\hat{y}_i$ (predicted tip amount)	Error $(y - \hat{y}_i)$	Squared Error $(y - \hat{y}_i)^2$
	x	y			
1	34	5	4.1505	0.8495	0.7217
2	108	17	14.9693	2.0307	4.1237
3	64	11	8.5365	2.4635	6.0688
4	88	8	12.0453	−4.0453	16.3645
5	99	14	13.6535	0.3465	0.1201
6	51	5	6.6359	−1.6359	2.6762
	$\bar{x} = 74$	$\bar{y} = 10$		SSE=	$\sum = 30.075$

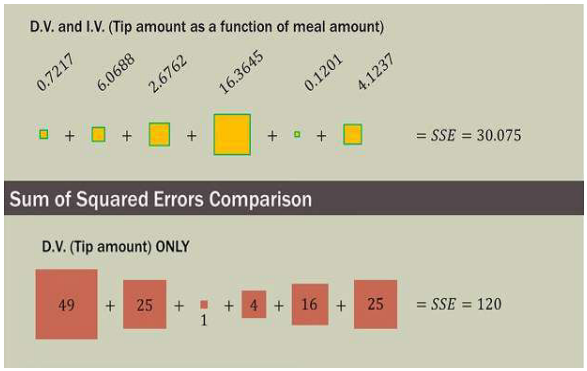
Regression Errors (Residuals)



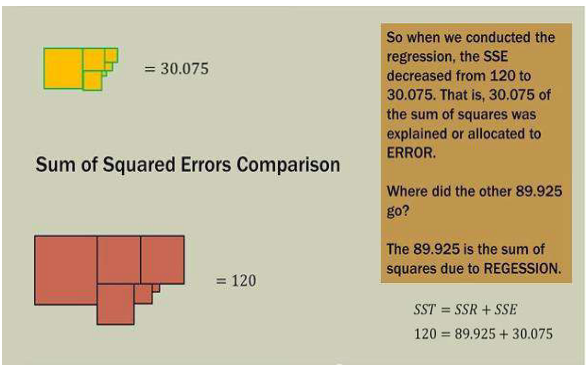
Squared Errors (Residuals)



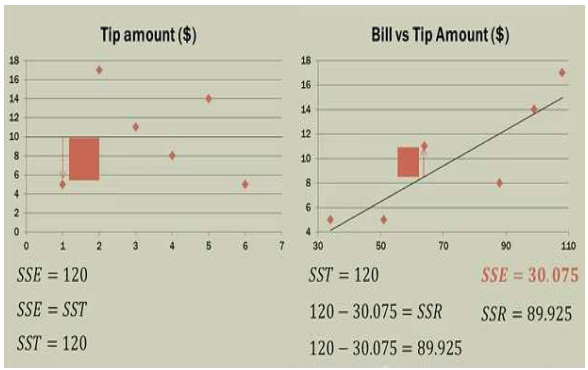
SSE Comparison



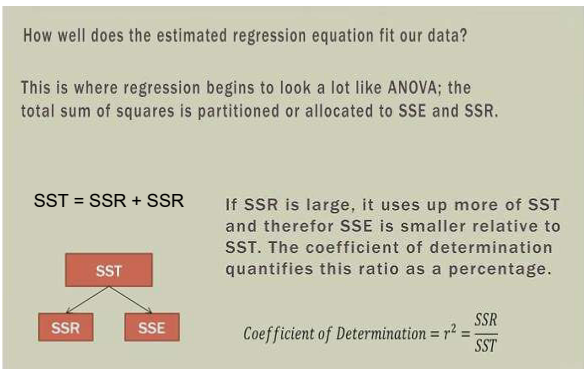
SSE Comparison



SSE Comparison



Coefficient of Determination



r^2 Interpretation

$$Coefficient\ of\ Determination = r^2 = \frac{SSR}{SST}$$

GOOD FIT!

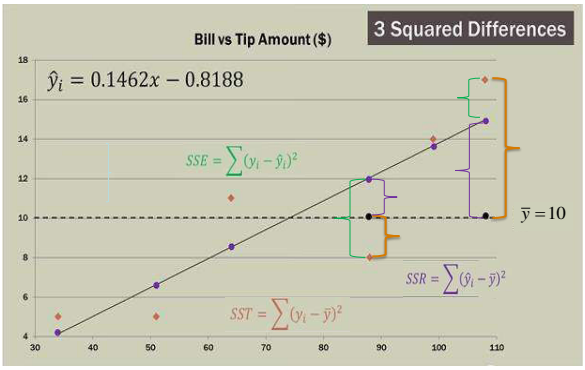
$$Coefficient\ of\ Determination = r^2 = \frac{89.925}{120}$$

$$Coefficient\ of\ Determination = r^2 = 0.7493\ or\ 74.93\%$$

We can conclude that 74.93% of the total sum of squares can be explained by using the estimated regression equation to predict the tip amount. The remainder is error.

$\hat{y}_i = 0.1462x - 0.8188$ Where x is the dollar amount of the bill.

SST = SSR + SSE



Summary

- Simple linear regression is really a **comparison of two models**
  - One is where the independent variable does not even exist
  - And the other uses the best fit regression line
- If there is only one variable, the best prediction for other values is the **mean** of the "dependent" variable
- The difference between the best-fit line and the observed value is called the **residual (or error)**
- The residuals are squared and then added together to generate **sum of squares (LITERALLY) residuals / error, SSE.**
- Simple linear regression is designed to find the best fitting line through the data that minimizes the SSE.
- How well does the estimated regression equation fit our data?
  - **The coefficient of determination**

Assumptions about Simple Regression Line

In order to use regression line, we have to make a few assumptions about the behavior of the data in a regression model.

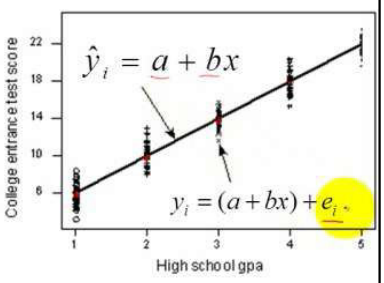
Example

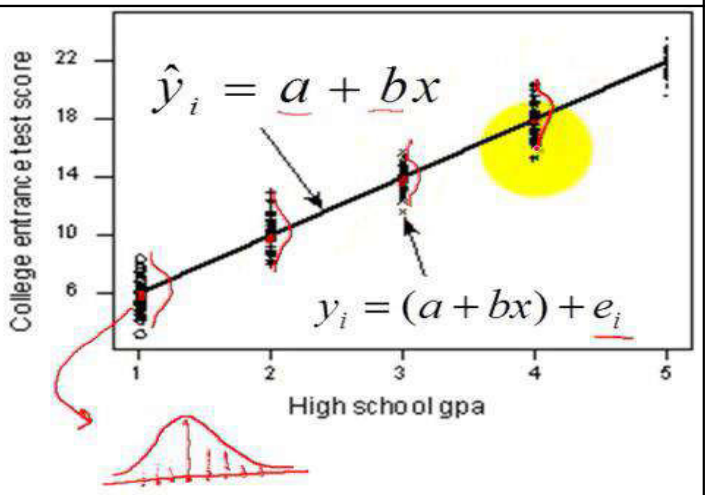
Let's focus for now just on those students who have a gpa of 1

Now, take the average college entrance test score for students with a gpa of 1.

And, similarly, take the average college entrance test score for students with a gpa of 2, 3, 4, and 5.

Connecting the dots — that is, the averages — you get a line, which we summarize by the formula





In order to use regression line, we have to make a few more assumptions about the behavior of the data in a regression setting.

Assumptions of Least Square line

- The relationship of y and x is in the form of a **Linear** function. (That is, it's shape is like a line, NOT a curve etc)

$y = a + bx + e$

- The errors terms (e_i, e_j) are **Independent** from each other. Also, "e" and "x" has no effect with each other.

Assumptions of Least Square line

- The errors e_i are **Normally distributed**



$\sum_{i=1}^{n} e_i = 0 \Rightarrow$ Mean of $e_i = 0$

Var of $e_i = \sigma^2$ = Constant

y_i	$\hat{y}_i$	e	$Var(e_i)$

- The errors e_i have **Equal Variances** (denoted σ^2) for all x values.

Summary

- We are now ready to summarize the four conditions that comprise "**the simple linear regression model**"

- The mean of the responses (y) is a **Linear function**
- The errors are **Independent**.
- The errors are **Normally distributed**.
- The errors have **Equal variances** (denoted σ^2) for all x values.
- **How to Remember** ➔ LINE

Properties of the Least square Regression Line

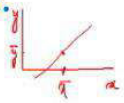
The regression line fitted by the Least Square method have some unique and useful properties

Properties of Least Square Regression Line

- The Least Square regression line always passes through the **center of the data.**

Average

$(\bar{x}, \bar{y})$



- The sum of deviations of the observed values of y from the LS regression line is always equal to **zero.**

$\sum e_i = (y_i - \hat{y}_i) = 0 \Rightarrow$

Properties of Least Square Regression Line

- The sum of the squares deviations of the observed values of y from the LS regression line is **minimum.**

$\sum e_i^2 = \sum (y_i - \hat{y}_i)^2 \Rightarrow$ Minimum

- The line fitted by LS method is **"best"** fit line.

LS estimator/estimate, → BLUE

Best Linear unbiased Estimator

The END

STA404
Regression & Correlation Analysis
Lesson 4

What is Correlation?

Introduction, basic concepts and properties

Correlation

The word Correlation is made of **Co-** (meaning "together"), and **Relation**

Definition:

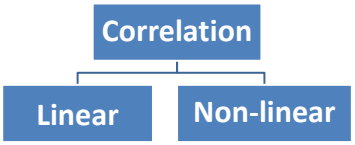
Two variables are said to be *correlated* if they tend to simultaneously vary together in some direction.

If the quantities(X,Y) vary in such a way that change in one variable corresponds to change in the other variable then the variables X and Y are correlated.

Daily life Examples of Correlation

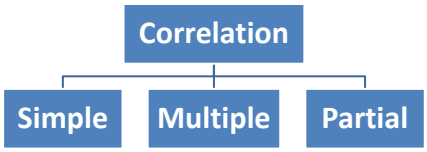
- Taller people have larger shoe sizes
- As the temperature goes up, ice cream sales also go up.
- As the slope of a hill increases, the amount of speed a walker reaches may decrease.
- The older a man gets, the less hair that he has.
- The more years of education you complete, the higher your earning potential will be.

Types of Correlation



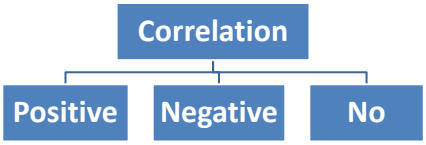
- When plotted on a graph it looks like a straight line
- When plotted on a graph it is NOT a straight line

Types of Correlation



- Involve two and one dependent variable
- One dependent and more than one independent variables
- One dependent variable and more than one independent variable but only one independent variable is considered and other independent variables are considered constant

Types of Correlation



- A positive correlation is a relationship between two variables where if one variable increases, the other one also increases. A positive correlation also exists if one decreases and the other also decreases.
- A negative correlation means that there is an inverse relationship between two variables - when one variable decreases, the other increases. The vice versa is a negative correlation too, in which one variable increases and the other decreases.
- Their movement is independent from each other.

Methods of Studying Correlation

The following are the important methods of ascertaining relationship between two variables.

- ❖ Scatter plot method
- ❖ Karl Pearson's Coefficient
- ❖ Spearman's Rank Correlation Coefficient

Scatter Plot Method:

The simplest device for studying correlation between two variables is a scatter plot.

How to measure correlation numerically?

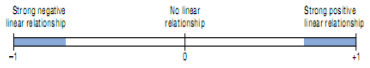
- Statisticians use a measure called the **correlation coefficient** to determine the strength of the linear relationship between two variables.
- There are several types of correlation coefficients.
- The one explained in this section is called the Pearson product moment correlation coefficient (PPMC),
- It was named after statistician Karl Pearson, who pioneered the research in this area.

Correlation Coefficient

- The correlation coefficient computed from the sample data measures the **strength and direction** of a linear relationship between two variables.
- The symbol for the sample correlation coefficient is r.
- The symbol for the population correlation coefficient is (Greek small letter "rho").

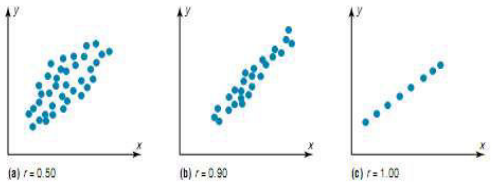
Correlation Coefficient – Important points

- The range of the correlation coefficient is from -1 to +1.



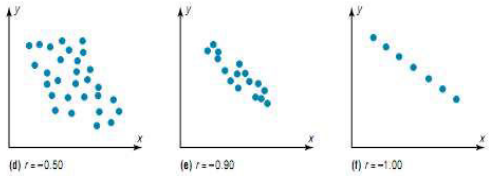
- If there is a strong positive linear relationship between the variables, the value of r will be close to 1.
- If there is strong negative linear relationship between the variables, the value of r will be close to1.
- When there is no linear relationship between the variables or only a weak relationship, the value of r will be close to 0.

Relationship between correlation coefficient and scatter plot



Notice that as the value of the correlation coefficient increases from 0 to 1 (parts *a, b, and c), data values become closer to an increasingly stronger relationship.*

Relationship between correlation coefficient and scatter plot



As the value of the correlation coefficient decreases from 0 to 1 (parts *d, e, and f), the data values also become closer to a straight line. Again this suggests a stronger relationship.*

Summary

- The word Correlation is made of Co- (meaning "together"), and Relation
- Correlation can be simple, multiple and partial
- It can be liner or nonlinear
- Positive and negative correlation
- Correlation coefficient is used to measure its numerical value
- Range of correlation coefficient is from -1 to +1

- The END -

STA404
Regression & Correlation Analysis
Lesson 4

How to calculate correlation coefficient?

Step by step procedure explained by an example and its interpretation

How to measure correlation numerically?

- Statisticians use a measure called the **correlation coefficient** to determine the strength of the linear relationship between two variables.
- There are several types of correlation coefficients.
- The one explained in this section is called the Pearson product moment correlation coefficient (PPMC),
- It was named after statistician Karl Pearson, who pioneered the research in this area.

How to calculate Correlation Coefficient - r

Statisticians use a measure called the **correlation coefficient** to determine the strength of the linear relationship between two variables.

Formula for the Correlation Coefficient *r*

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n(\Sigma x^2) - (\Sigma x)^2][n(\Sigma y^2) - (\Sigma y)^2]}}$$

where n is the number of data pairs.

- There are no units associated with *r,*
- *the value of r will remain unchanged if the x and y values are switched.*

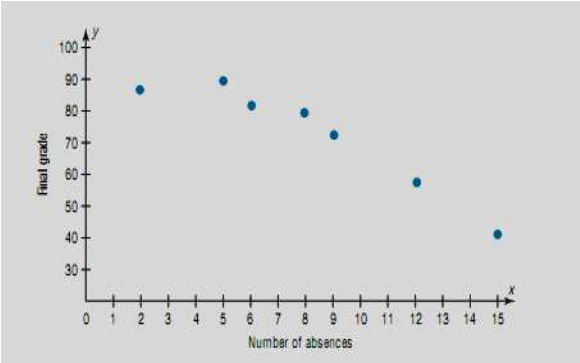
Example

Absences and Final Grades

Construct a scatter plot for the data obtained in a study on the number of absences and the final grades of seven randomly selected students from a statistics class. The data are shown here.

Student	Number of absences x	Final grade y (%)
A	6	82
B	2	86
C	15	43
D	9	74
E	12	58
F	5	90
G	8	78

Scatter plot



Calculating r

Solution

Step 1 Make a table.

Step 2 Find the values of xy, x^2, and y^2; place these values in the corresponding columns of the table.

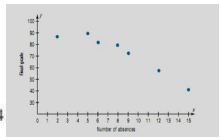
Student	Number of absences x	Final grade y (%)	xy	x^2	y^2
A	6	82	492	36	6,724
B	2	86	172	4	7,396
C	15	43	645	225	1,849
D	9	74	666	81	5,476
E	12	58	696	144	3,364
F	5	90	450	25	8,100
G	8	78	624	64	6,084
	$\Sigma x = 57$	$\Sigma y = 511$	$\Sigma xy = 3745$	$\Sigma x^2 = 579$	$\Sigma y^2 = 38{,}993$

Interpreting r

Step 3 Substitute in the formula and solve for r.

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n(\Sigma x^2) - (\Sigma x)^2][n(\Sigma y^2) - (\Sigma y)^2]}}$$

$$= \frac{(7)(3745) - (57)(511)}{\sqrt{[(7)(579) - (57)^2][(7)(38{,}993) - (511)^2]}} = -0.944$$



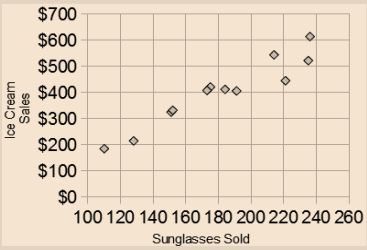
What it shows (Interpretation)?
The value of *r suggests a strong negative relationship between a student's final grade and the number of absences a student has. That is, the more absences a student has, the lower is his or her grade.*

An important point …

"Correlation Is Not Causation" ... which says that a correlation does **not** mean that one thing causes the other (there could be other reasons the data has a good correlation).

Example: Sunglasses vs Ice cream

Our Ice Cream shop finds how many sunglasses were sold by a big store for each day and compares them to their ice cream sales:



The correlation between Sunglasses and Ice Cream sales is high

Does this mean that sunglasses make people want ice cream?

Summary

"Correlation coefficient" is one of the method to measure it.

- Range of correlation coefficient is from -1 to +1
- Formula for calculation
- Correlation is NOT causation

- The END -

STA404
Regression & Correlation Analysis
Lesson 5

Rank Correlation

Why we need it and how to measure it, explained with an example

Correlation

Methods of Studying Correlation:

The following are the important methods of ascertaining between two variables.

- ❖ Scatter diagram method
- ❖ Karl Pearson's Simple Correlation Coefficient
- ❖ Spearman's Rank Correlation Coefficient

Limitations …

- By applying Scatter diagram method we can get an idea about the direction of correlation and also whether it is high or low.
- But we cannot establish the exact degree of correlation between the variables as it is possible by applying mathematical methods.

- The Karl Pearson's method is based on the assumption that the population being studied is **normally distributed.** When it is known that the population is not normal or when the shape the distribution is not known, there is need for a measure of correlation that involves no assumption about the parameter of the population.

So Why Rank Correlation????

- **NO ASSUMPTION NEEDED**: It is possible to avoid making any assumptions about the populations being studied by ranking the observations according to size and basing the calculations on the ranks rather than upon the original observations.
- **WHEN RANKED DATA:** When the given data is in the form of ranks, we have no other choice. It does not matter which way the items are ranked, item number one may be the largest or it may be the smallest. Using ranks rather than actual observations gives the coefficient of rank correlations.
- **DEVELOPED BY:** This method of finding out co variability or the lack of it between two variables was developed by the British Psychologist Charles Edward Spearman in 1904.

Rank Correlation Coefficient (formula)

It studies the relationships between different rankings on the same set of items. It deals with measuring correspondence between two rankings, and assessing the significance of this correspondence. Spearman's correlation coefficient is defined as:

$$r_s = 1 - \frac{6\Sigma d^2}{n(n^2-1)}$$

Where

r_s = Coefficient of rank correlation
d = difference
n=no .of data

Features

The rank method has principal uses:

- The **range** of rank correlation is -1 to +1
- The rank correlation can be **interpreted** in the same way as Karl Pearson's correlation coefficient.
- Rank correlation coefficient is a distribution-free measure since no strict **assumption** is made about the population from which it is drawn.
- Spearman's formula is the only formulae available to find the correlation between ranks (qualitative characters.)

Types of Rank Methods

Ranked data	• Where ranks are given
NOT ranked data	• Where ranks are not given
Ranks are tied	• Where repeated ranks occur

Example: ICC Ranking

- The ICC ranking for One Day International (ODI) and Test match is given in the table.
- Test whether there is correlation between both performances

Team	Test Rank	ODI Rank
Australia	1	1
India	2	3
South Africa	3	2
Sri Lanka	4	7
England	5	6
Pakistan	6	4
New Zealand	7	5
West Indies	8	8
Bangladesh	9	9

Calculation Steps

Step1 : Make the table

Step 2: Fill the data with d and d^2

Step 3: Plug the values in the formula

Team	Test Rank	ODI Rank	d = Difference of ranks	d^2
Australia	1	1	0	0
India	2	3	-1	1
South Africa	3	2	1	1
Sri Lanka	4	7	-3	9
England	5	6	-1	1
Pakistan	6	4	2	4
New Zealand	7	5	2	4
West Indies	8	8	0	0
Bangladesh	9	9	0	0
TOTAL	-	-	-	20

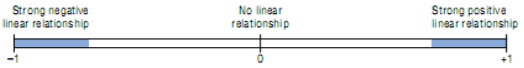
Calculation Steps

- Putting the values in the formula

$$r_s = 1 - \frac{6\Sigma d_i^2}{n(n^2-1)}$$
$$= 1 - \frac{6(20)}{9(81-1)}$$
$$= 1 - \frac{120}{720}$$
$$= 0.8333$$

Interpreting Results

- Rank Correlation = 0.8333



- We see that its shows a strong positive correlation. Which suggest that the performance in ODI and test match go side by side. The team with good ranking in ODI will also enjoy good ranking in test match and vice versa.

Summary

- In many real life situations, we are unable to get actual data, rather we have ranked data
- Karl Pearson Coefficient formula - Normality assumption needed
- Use Spearman's' Rank Correlation $r_s = 1 - \frac{6\Sigma d_i^2}{n(n^2-1)}$
- Range is same -1 to +1
- Interpret result in same manner

- The END -

STA404
Regression & Correlation Analysis
Lesson 6

Multiple Linear Regression Analysis

Why move to multiple linear regression?

Why one variable is NOT enough?

- Yield of crops depends on … ???
  - Fertility of land
  - Fertilizer applied
  - Rainfall
  - Quality of seeds
- Sale of a company depends on … ???
  - Price
  - Demand
  - Advertisement
  -

Example: Concept

- Suppose a nursing instructor wishes to see whether there is a relationship between a student's grade point average **(GPA), age,** and score on the board examination. The two independent variables are GPA (denoted by x_1) and age (denoted by x_2).
- The instructor will collect the **data for all three variables** for a sample of nursing students.
- **Rather than** conduct two separate simple regression studies, one using the GPA and state board scores and another using ages and state board scores, the instructor can conduct one study using multiple regression analysis with two independent variables— GPA and ages—and one dependent variable—state board scores.

Example: Looks like

For example, the nursing instructor wishes to see whether a student's grade point average and age are related to the student's score on the state board nursing exam. She selects five students and obtains the following data.

Student	GPA x_1	Age x_2	State board score y
A	3.2	22	550
B	2.7	27	570
C	2.5	24	525
D	3.4	28	670
E	2.2	23	490

The multiple regression equation obtained from the data is

$$y' = -44.81 + 87.64x_1 + 14.533x_2$$

Simple >>> Multiple Regression

Simple Linear Regression

- In simple linear regression, the regression equation contains one independent variable *x and one dependent variable y and is written as*

$$\hat{Y} = a + bX$$

Multiple Linear Regression

- A multiple regression equation with **two independent** variables (x_1 and x_2) and one dependent variable has the form:

$$\hat{Y} = a + b_1X_1 + b_2X_2$$

General form

- A multiple regression equation with **three independent** variables (x_1, x_2 and x_3) and one dependent variable has the form:

$$\hat{Y} = a + b_1X_1 + b_2X_2 + b_3X_3$$

- The **general form** of the multiple regression equation with ***k independent*** *variables is*

$$\hat{Y} = a + b_1X_1 + b_2X_2 + \cdots + b_kX_k$$

- *Y is called DEPENDENT variable*
- The *x's are the independent variables.*
- The *b's are called partial regression coefficients.*
- *K is number of independent variables*

Assumptions

- The assumption for Linear Multiple Regression line are similar to those Simple Linear Regression , as already discussed. In a summary form:

- The mean of the responses is a **Linear function**
- The errors are **Independent**.
- The errors are **Normally distributed**.
- The errors have **Equal variances** (denoted σ^2) for all *x* values.

- And For Multiple: we add this:
- We also consider that the variables X_1 and X_2 are NOT correlated with each other.

Summary

- Multiple: Because we are dealing with more than one independent variables
- With TWO variables it looks like this:

$$\hat{Y} = a + b_1X_1 + b_2X_2$$

- Coefficient are called partial coefficients
- Similar assumption like the simple liner regression

STA404
Regression & Correlation Analysis
Lesson 6

Multiple Regression Analysis

Estimating multiple regression by solving the NORMAL equations simultaneously

Example

A researcher wants to know the earning of restaurants, using two independent variables: The number of employees and restaurant area/size. He collected the following data.

Income Y	Area X_1	Number of employees X_2
30	10	15
22	5	8
16	10	12
7	3	7
14	2	10

Calculate the estimated multiple linear regression equation for the above data.

Normal Equations

$$\hat{Y} = a + b_1X_1 + b_2X_2$$

- $\sum Y = na + b_1\sum X_1 + b_2\sum X_2 \longrightarrow ①$
- $\sum X_1Y = a\sum X_1 + b_1\sum X_1^2 + b_2\sum X_1X_2 \longrightarrow ②$
- $\sum X_2Y = a\sum X_2 + b_1\sum X_1X_2 + b_2\sum X_2^2 \longrightarrow ③$

Calculations Table

	Given Data			Required Calculations				
	Y	X_1	X_2	X_1^2	X_2^2	X_1Y	X_2Y	X_1X_2
	30	10	15	100	225	300	450	150
	22	5	8	25	64	110	176	40
	16	10	12	100	144	160	192	120
	7	3	7	9	49	21	49	21
	14	2	10	4	100	28	140	20
Total	89	30	52	238	582	619	1007	351

How to Solve Normal Equations?

- *First, Putt the relevant values in the equations,*
- $\sum Y = na + b_1 \sum X_1 + b_2 \sum X_2$
- $5a + 30b_1 + 52b_2 = 89 \longrightarrow ①$
- $\sum X_1 Y = a \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1 X_2$
- $30a + 238b_1 + 351b_2 = 619 \longrightarrow ②$
- $\sum X_2 Y = a \sum X_2 + b_1 \sum X_1 X_2 + b_2 \sum X_2^2$
- $52a + 351b_1 + 582b_2 = 1007 \longrightarrow ③$

How to Solve Normal Equations?

- $5a + 30b_1 + 52b_2 = 89 \longrightarrow ①$
- $30a + 238b_1 + 351b_2 = 619 \longrightarrow ②$
- $52a + 351b_1 + 582b_2 = 1007 \longrightarrow ③$

Solving ① and ② simultaneously…

$$5a + 30b_1 + 52b_2 = 89 \longrightarrow ①$$

- Multiply the above equation ① with 6:

$$30a + 180b_1 + 312b_2 = 534$$

- Now subtract this equation from equation ②, we get

$$\begin{array}{r} 30a + 238b_1 + 351b_2 = 619 \\ 30a + 180b_1 + 312b_2 = 534 \\ \hline 0 + 58b_1 + 39b_2 = 85 \end{array}$$

$$58b_1 + 39b_2 = 85 \longrightarrow ④$$

Solving ② and ③ simultaneously…

$$30a + 238b_1 + 351b_2 = 619 \longrightarrow ②$$

- Multiply it with 52, we get below
- $1560a + 12376b_1 + 18252b_2 = 32188 - -say \longrightarrow ⑤$

$$52a + 351b_1 + 582b_2 = 1007 \longrightarrow ③$$

- Multiply it with 30, we get below

$1560a + 10530b_1 + 17460b_2 = 30210 - -say \longrightarrow ⑥$

Solving ② and ③ simultaneously…

- *Now subtract* ⑤ *and* ⑥

$$\begin{array}{r} 1560a + 12376b_1 + 18252b_2 = 32188 \\ 1560a + 10530b_1 + 17460b_2 = 30210 \\ \hline 0 + 1846b_1 + 792b_2 = 1978 \end{array}$$

$$1846b_1 + 792b_2 = 1978 \longrightarrow ⑦$$

Now Solving TWO new Equations…

- $5a + 30b_1 + 52b_2 = 89 \longrightarrow ①$
- $30a + 238b_1 + 351b_2 = 619 \longrightarrow ②$
- $52a + 351b_1 + 582b_2 = 1007 \longrightarrow ③$
- $58b_1 + 39b_2 = 85 \longrightarrow ④$
- $1846b_1 + 792b_2 = 1978 \longrightarrow ⑦$

==========

- *Multiply* ④ with 1846 *and* ⑦ *with 58 and subtract the new equations from each other*

$$\begin{array}{r} 107068b_1 + 71994b_2 = 156910 \\ 107068b_1 + 45936b_2 = 114724 \\ \hline 0 + 26058b_2 = 42186 \end{array}$$

$$\boxed{b_2 = 1.62}$$

Getting value of $b1$…

- *Now put value of* b_2 *in any of following equation and solve it to value of* b_1

$$107068b_1 + 71994(1.62) = 156910$$
$$107068b_1 = 40279.72$$

$$\boxed{b_1 = 0.38}$$

Getting value of a…

Now we put newly computed value of $b_1$and b_2 in equation

$$5a + 30b_1 + 52b_2 = 89 \longrightarrow ①$$
$$5a + 30(0.38) + 52(1.62) = 89 \longrightarrow ①$$
$$5a = -6.64$$

$$\boxed{a = -1.33}$$

Final Equation

Now put values of $\boxed{a = -1.33}$, $\boxed{b_1 = 0.38}$ $\boxed{b_2 = 1.62}$ in equation

$$\hat{Y} = a + b_1X_1 + b_2X_2$$

$$\hat{Y} = -1.33 + 0.38\ X_1 + 1.62\ X_2$$

$$\hat{Y} = -1.33 + 0.38\ (Area) + 1.62\ (No.\ of\ Employee)$$

Interpretation

$$\hat{Y} = -1.33 + 0.38\ X_1 + 1.62\ X_2$$

$$\hat{Y} = -1.33 + 0.38\ (Area) + 1.62\ (No.\ of\ Employee)$$

What is the effect of variables on income?

In this case, for each unit of change in the area of floor (X_1), there is a change of 0.38 units in the income with the number of employees being held constant.

And for each unit of change in X_2 (the number of employees), there is a change of 1.62 units in the income with the floor area held constant.

Interpretation

$$\hat{Y} = -1.33 + 0.38\ X_1 + 1.62\ X_2$$

$$\hat{Y} = -1.33 + 0.38\ (Area) + 1.62\ (No.\ of\ Employee)$$

What is the effect of variables on income?

In this case, for each unit of change in the area of floor (X_1), there is a change of 0.38 units in the income with the number of employees being held constant.

And for each unit of change in X_2 (the number of employees), there is a change of 1.62 units in the income with the floor area held constant.

Summary

- Take multiple model
- Make the 3 normal equations
- Calculate the required values in the table
- Reduce the 3 normal equations to the 2 equations
- Solve the new two equations to get unknown constants, step by step.
- Put the constants in the model

STA404
Regression & Correlation Analysis
Lesson 6

Multiple Regression Analysis

Estimating multiple regression by using the FORMULA and deviation method

Example

A researcher wants to know the earning of restaurants, using two independent variables: The number of employees and restaurant area/size. He collected the following data.

Income Y	Area X_1	Number of employees X_2
30	10	15
22	5	8
16	10	12
7	3	7
14	2	10

Calculate the estimated multiple linear regression equation for the above data.

Formula for b_1 and b_2 in (deviation form)

Instead of solving the 3 normal equations simultaneously, which is tedious work, alternatively we can use **FORMULA** to find the coefficient values.

$$b_1 = \frac{(\sum x_1 y)(\sum x_2{}^2) - (\sum x_2 y)(\sum x_1 x_2)}{(\sum x_1{}^2)(\sum x_2{}^2) - (\sum x_1 x_2)^2}$$

$$b_2 = \frac{(\sum x_2 y)(\sum x_1{}^2) - (\sum x_1 y)(\sum x_1 x_2)}{(\sum x_1{}^2)(\sum x_2{}^2) - (\sum x_1 x_2)^2}$$

$$a = \bar{Y} - b_1\bar{X}_1 - b_2\bar{X}_2$$

Convert data into *deviation form*

	Original data			*Data in DEVIATION form*		
	Y	X_1	X_2	$y = Y - \bar{Y}$		
	30	10	15	12.2		
	22	5	8	4.2		
	16	10	12	-1.8		
	7	3	7	-10.8		
	14	2	10	-3.8		
Total	89	30	52			
Mean	17.8	6	10.4			

Convert data into *deviation form*

	Original data			*Data in DEVIATION form*		
	Y	X_1	X_2	$y = Y - \bar{Y}$	$x_1 = X_1 - \bar{X}$	$x_2 = X_2 - \bar{X}$
	30	10	15	12.2	4	4.6
	22	5	8	4.2	-1	-2.4
	16	10	12	-1.8	4	1.6
	7	3	7	-10.8	-3	-3.4
	14	2	10	-3.8	-4	-0.4
Total	89	30	52	0	0	0
Mean	17.8	6	10.4			

Deviation form > CheckPoint

Further Computation on deviation form data

	Data in DEVIATION form			*Required calculations for formula*				
	y	x_1	x_2	x_1y	x_2y	x_1x_2	x_1^2	x_2^2
1	12.2	4	4.6	48.8	56.12	18.4	16	21.16
2	4.2	-1	-2.4	-4.2	-10.08	2.4	1	5.76
3	-1.8	4	1.6	-7.2	-2.88	6.4	16	2.56
4	-10.8	-3	-3.4	32.4	36.72	10.2	9	11.56
5	-3.8	-4	-0.4	15.2	1.52	1.6	16	0.16
Total	0	0	0	85	81.4	39	58	41.2

Calculating value of b_1

$$b_1 = \frac{(\sum x_1 y)(\sum x_2{}^2) - (\sum x_2 y)(\sum x_1 x_2)}{(\sum x_1{}^2)(\sum x_2{}^2) - (\sum x_1 x_2)^2}$$

$$b_1 = \frac{(85)(41.2) - (81.4)(39)}{(58)(41.2) - (39)^2}$$

$$b_1 = \frac{327.4}{868.6} = 0.38$$

Calculating value of b_2

$$b_2 = \frac{(\sum x_2 y)(\sum x_1{}^2) - (\sum x_1 y)(\sum x_1 x_2)}{(\sum x_1{}^2)(\sum x_2{}^2) - (\sum x_1 x_2)^2}$$

$$b_2 = \frac{(81.4)(58) - (85)(39)}{(58)(41.2) - (39)^2}$$

$$b_2 = \frac{1406.2}{868.6} = 1.62$$

Calculating value of a

$$a = \bar{Y} - b_1\bar{X}_1 - b_2\bar{X}_2$$

$$a = 17.8 - (0.38)(6) - (1.62)(10.4)$$

$$a = -1.33$$

Final Equation

- Hence the desired multiple linear regression equation is:

$$\hat{Y} = a + b_1X_1 + b_2X_2$$

$$\hat{Y} = -1.33 + 0.38\,X_1 + 1.62\,X_2$$

$$\hat{Y} = -1.33 + 0.38(Area) + 1.62(No.\,of\ Employee)$$

Interpretation

$$\hat{Y} = -1.33 + 0.38\,X_1 + 1.62\,X_2$$

$$\hat{Y} = -1.33 + 0.38(Area) + 1.62(No.\,of\ Employee)$$

What is the effect of variables on income?

In this case, for each unit of change in the area of floor (X_1), there is a change of 0.38 units in the income with the number of employees being held constant.

And for each unit of change in X_2 (the number of employees), there is a change of 1.62 units in the income with the floor area held constant.

Summary

- Instead of solving the 3 normal equations simultaneously, which is tedious work, alternatively we can use **FORMULA** to find the coefficient values.
- Formula is designed to apply on the data in ***deviation form***: It will eliminate some values to make computations simple.
- Insert the constants in regression model and interpret the results.

STA404
Regression & Correlation Analysis
Lesson 7-1

Multiple Correlation

Finding correlation between one variable and other group of variables Simultaneously

Some Change in Notations

$$Y = a + b_1X_1 + b_2X_2$$

$$X_1 = b_{1.23} + b_{12.3}X_2 + b_{13.2}X_3$$

Variable 1	Variable 2	Variable 3
Independent	Dependent	Dependent
X_1	X_2	X_3
.	.	.
.	.	.

- Variables are numbered 1, 2 and 3 by subscript
- Subscript 1 shows ***dependent*** and 2 and 3 show ***independent*** variables

Some Change in Notations

$Y = a + b_1X_1 + b_2X$

Which OTHER variable is included in equation

$b_{12.3}X_2$

Shows DEPENDET variable

Coefficients is attached with which variable

$$X_1 = b_{1.23} + b_{12.3}X_2 + b_{13.2}X_3$$

Multiple Correlation

- It study the effects of ALL the independent variables ***simultaneously*** on a dependent variable.



- For example, the correlation co-efficient between the yield of rice (X_1) AND **ALL other** variables like type of seeds (X_2), fertilizer (X_3), rainfall (X_4), humidity (X_5) is the multiple correlation co-efficient.
- This **multiple correlation coefficient is symbolized by $R_{1.2345}$**
- It can be can be computed by using the values of the individual correlation coefficients.

Multiple Correlation Coefficient - Formula

The coefficient of multiple correlation between X_1 and ***the variables X_2 and X_3 combined***, is denoted by $R_{1.23}$ and calculated by:

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{23}^2}}$$

Variable 1	Variable 2	Variable 3
Independent	Dependent	Dependent
X_1	X_2	X_3
.	.	.
.	.	.

Where ;

$r_{12} = r_{21}$ = *Simple correlation between variable 1 and 2*

$r_{13} = r_{31}$ = *Simple correlation between variable 1 and 3*

$r_{23} = r_{32}$ = *Simple correlation between variable 2 and 3*

Multiple Correlation Coefficient – Other Forms of Formula

The coefficient of multiple correlation between X_1 and ***the variables X_2 and X_3 combined***, is denoted by $R_{1.23}$ and calculated by:

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$$

Variable 1	Variable 2	Variable 3
Independent	Dependent	Dependent
X_1	X_2	X_3
.	.	.
.	.	.

Similarly,

$R_{2.13} = ?$

$R_{3.12} = ?$

Swapping Subscript

The coefficient of multiple correlation between X_1 and ***the variables X_2 and X_3 combined***, is denoted by $R_{1.23}$ and calculated by:

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$$

$$R_{1.32} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$$

Similarly,

$R_{2.13} = R_{2.31}$

$R_{3.12} = R_{3.21}$

Multiple Correlation – Main Points

- ***The value of R can range from 0 to 1;***
- *R can never be negative.*
- *The closer to 1, the stronger the relationship;*
- *The closer to 0, the weaker the relationship.*
- The value of *R takes into account all the independent variables*

Example: Calculating Multiple Correlation

For multiple regression model the simple correlation coefficients are $r_{12} = 0.492$, $r_{13} = 0.927$, $r_{23} = 0.758$. Find all the multiple correlation coefficients.

SOLUTION:

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$$

$$R_{1.23} = \sqrt{\frac{(0.492)^2 + (0.927)^2 - 2(0.492)(0.927)(0.758)}{1 - (0.758)^2}}$$

$$R_{1.23} = \sqrt{\frac{0.41}{0.4254}}$$

$$R_{1.23} = 0.4024$$

Example: Can you do yourself?

For multiple regression model the simple correlation coefficients are $r_{12} = 0.492$, $r_{13} = 0.927$, $r_{23} = 0.758$. Find all the multiple correlation coefficients.

SOLUTION:

$$R_{2.13} = \sqrt{\frac{r_{21}^2 + r_{23}^2 - 2r_{21}r_{23}r_{13}}{1 - r_{13}^2}}$$

$$R_{3.12} = _____$$

Summary

- For multiple independent variables, we use multiple correlation
- It is correlation between one variable with others multiple/group of variables
- It is denoted by like: $R_{1.23}$
- Using simple correlations, we can find out the multiple correlation
- Its range is from 0 to +1

The END

STA404
Regression & Correlation Analysis
Lesson 7-2

Partial Correlation

Finding correlation between any two variables excluding other variables

Some Change in Notations

$$Y = a + b_1X_1 + b_2X_2$$

$$X_1 = b_{1.23} + b_{12.3}X_2 + b_{13.2}X_3$$

- Variables are numbered 1, 2 and 3 by subscript
- Subscript 1 shows ***dependent*** and 2 and 3 show ***Inependent*** variables

Some Change in Notations

$Y = a + b_1X_1 + b_2X_2$

Which OTHER variable is included in equation

$b_{12.3}$

Shows DEPENDET variable

Coefficients is attached with which variable

$X_1 = b_{1.23} + b_{12.3}X_2 + b_{13.2}X_3$

Partial Correlation

- Partial correlation analysis involves studying the linear relationship between two variables **after excluding** the effect of one or more independent factors.
- Simple correlation does not prove to be a good technique especially when we are dealing with multiple variables. In order to get a correct picture of the relationship between two variables, we should *first eliminate the influence of other variables.*



- For example, the correlation between the yield of rice (X_1) AND the quantity of seeds (X_2), **EXCLUDING** the effect of fertilizer (X_3), rainfall (X_4), humidity (X_5) is partial correlation.
- This **partial correlation coefficient is denoted by (small) $r_{12.345}$**

Partial Correlation Coefficient - Formula

The correlation co-efficient between two variables X_1 and X_2, studied partially after eliminating the influence of the third variable X_3 from both of them, is the partial correlation co-efficient $r_{12.3}$ and can be calculated by:

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{1-r_{13}^2}\sqrt{1-r_{23}^2}}$$

Variable 1	Variable 2	Variable 3
Dependent	Independent	Independent
X_1	X_2	X_3
.	.	.
.	.	.

Where ;

$r_{12} = r_{21}$ = *Simple correlation between variable 1 and 2*

$r_{13} = r_{31}$ = *Simple correlation between variable 1 and 3*

$r_{23} = r_{32}$ = *Simple correlation between variable 2 and 3*

Partial Correlation Coefficient – Other Forms

The partial correlation coefficient between X_1 and X_2 by excluding the influence of X_3 is denoted by $r_{12.3}$ and calculated by:

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{1-r_{13}^2}\sqrt{1-r_{23}^2}}$$

Variable 1	Variable 2	Variable 3
Dependent	Independent	Independent
X_1	X_2	X_3
.	.	.
.	.	.

Similarly,

$r_{23.1} = ?$

$r_{13.2} = ?$

Swapping Subscript

The partial correlation coefficient between X_1 and X_2 by excluding the influence of X_3 is denoted by $r_{12.3}$ and calculated by:

$$r_{12.3} = r_{21.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{1-r_{13}^2}\sqrt{1-r_{23}^2}}$$

Similarly,

$r_{23.1} = r_{32.1} = ?$

$r_{13.2} = r_{31.2} = ?$

Partial Correlation – Main Points

- *The value of r can range from -1 to +1;*
- *The closer to 1, the stronger the relationship;*
- *The closer to 0, the weaker the relationship.*
- Its calculation is based on the simple correlation co-efficient.
- *It excludes (= held constant, eliminate, removing) the influence of other variables.*

Example: Calculating Partial Correlation

For multiple regression model the simple correlation coefficients are $r_{12} = 0.492$, $r_{13} = 0.927$, $r_{23} = 0.758$. Find all the **partial correlation** coefficients.

SOLUTION:

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{1-r_{13}^2}\sqrt{1-r_{23}^2}}$$

$$r_{12.3} = \frac{0.492 - (0.927)(0.758)}{\sqrt{1-(0.927)^2}\sqrt{1-(0.758)^2}}$$

$$r_{12.3} = \frac{-0.2107}{\sqrt{0.1407}\sqrt{0.4254}}$$

$$r_{12.3} = -0.86$$

Can you do yourself?

For multiple regression model the simple correlation coefficients are $r_{12} = 0.492$, $r_{13} = 0.927$, $r_{23} = 0.758$. Find ALL the **partial correlation** coefficients.

SOLUTION:

$$r_{23.1} = \frac{r_{23} - r_{21}r_{31}}{\sqrt{1-r_{21}^2}\sqrt{1-r_{31}^2}}$$

$r_{13.2} =$ _____

Summary

- **A partial correlation makes it possible to remove the effect of a third variable**
- It is denoted by $r_{12.3}$
- We use simple correlations to find out the partial correlation
- Like simple correlation, its range is from **-1 to +1**
- If multiple and partial correlation are studied together, a very useful analysis of the relationship between the different variables is possible.

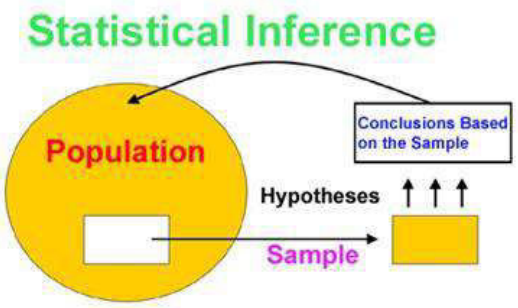
The END

STA404
Regression & Correlation Analysis
Lesson 8-1

Statistical Inference about Regression

Explanation of the hypothesis testing procedure for simple linear regression coefficient **β**

What is Statistical inference?



Difference between Population & Sample Regression

- Population Regression

$$y = \alpha + \beta x + \varepsilon$$

- Sample Regression

$$\hat{y} = a + bx + e$$

Hypothesis Testing – General Procedure

Procedure Table

Solving Hypothesis-Testing Problems (Traditional Method)

Step 1	State the hypotheses and identify the claim.
Step 2	Find the critical value(s) from the appropriate table
Step 3	Compute the test value.
Step 4	Make the decision to reject or not reject the null hypothesis.
Step 5	Summarize the results.

Testing Hypothesis – Procedure about Regression Coefficient β

Step 1

Stat the hypothesis.

$H_0: \beta = 0$ and $H_1: \beta \neq 0$
$H_0: \beta \leq 0$ and $H_1: \beta > 0$
$H_0: \beta \geq 0$ and $H_1: \beta < 0$

Step 2

Decide the level of significance.
Here, we set the value of $\alpha = 5\% = 0.05$

General formula format

Many hypotheses are tested using a statistical test based on the following general formula:

$$Test\ Value = \frac{(Observed\ value) - (Expected\ value)}{Standard\ error}$$

The observed value is the "statistic" (such as the mean) that is computed from the **sample data.**

The expected value is the "parameter" (such as the mean) that you would expect to obtain if the null hypothesis were true. In other words, the hypothesized value.

The denominator is the standard error of the statistic being tested (in this case, the standard error of the mean).

$$t = \frac{b - \beta_0}{S.E(b)}$$

It is t-test. It depends on *Degree of Freedom* = df = n-2

Required calculations

Step 4

Know and do the required calculations and find the test value.

$$t = \frac{b-\beta_0}{S.E(b)}$$

$$b = \frac{n\sum XY - (\sum X)(\sum Y)}{n\sum X^2 - (\sum X)^2}$$

$$a = \bar{Y} - b\bar{X}$$

$$s_{YX} = \sqrt{\frac{\sum(Y-\hat{Y})^2}{n-2}} = \sqrt{\frac{\sum Y^2 - a\sum Y - b\sum XY}{n-2}}$$

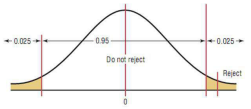
$$s_b = \frac{s_{YX}}{\sqrt{\sum(X-\bar{X})^2}}$$

Critical Value, Comparison and Conclusion

Step 5

- Use t-table to find the critical values.
- Plot these values on bell shaped curve.
- Compare calculated value with table value
- Conclude and summarize the result





Testing Hypothesis – Example

- For the Given data:
- Estimate the regression line from the following data of height (X) and Weight (Y)
- Test the hypothesis that the height and weight are independent.

Height (X)	Weight(Y)
60	110
60	135
60	120
62	120
62	140
62	130
62	135
64	150
64	145
70	170
70	185
70	160

Testing Hypothesis – Example

Step 1

Stat the hypothesis.

$H_0: \beta = 0$ (There is NO slop. There is NO linear relationship.)

$H_1: \beta \neq 0$ (There exist linear relationship)

Step 2

We set the level of significance (alpha) $\alpha = 0.05$

Step 3

The test-statistic (formula) to be used is:

$$t = \frac{b-\beta_0}{S.E(b)}$$

It follows t-distribution with degree of freedom ***df = n-2***

Computation

X	Y	XX	YY	XY
60	110	3600	12100	6600
60	135	3600	18225	8100
60	120	3600	14400	7200
62	120	3844	14400	7440
62	140	3844	19600	8680
62	130	3844	16900	8060
62	135	3844	18225	8370
64	150	4096	22500	9600
64	145	4096	21025	9280
70	170	4900	28900	11900
70	185	4900	34225	12950
70	160	4900	25600	11200
766	1700	49068	246100	109380

What we need to calculate?

Step 4

Do necessary computations from data and solve the formula using these values.

$$t = \frac{b-\beta_0}{s_b}$$

- $s_b = \frac{s_{YX}}{\sqrt{\Sigma(X-\bar{X})^2}}$ $\qquad b = \frac{n\Sigma XY - (\Sigma X)(\Sigma Y)}{n\Sigma X^2 - (\Sigma X)^2}$

$$s_{YX} = \sqrt{\frac{\Sigma(Y-\hat{Y})^2}{n-2}} = \sqrt{\frac{\Sigma Y^2 - a\Sigma Y - b\Sigma XY}{n-2}}$$

$$a = \bar{Y} - b\bar{X}$$

$$\Sigma(X-\bar{X})^2 = \Sigma X^2 - \frac{(\Sigma X)^2}{n}$$

Let's do calculations

Step 4

$$b = \frac{n\Sigma XY - (\Sigma X)(\Sigma Y)}{n\Sigma X^2 - (\Sigma X)^2} = _____ = _____ =$$

$$a = \bar{Y} - b\bar{X} = \qquad\qquad =$$

$$\Sigma(Y-\hat{Y})^2 = \Sigma Y^2 - a\Sigma Y - b\Sigma XY$$

$$\Sigma(X-\bar{X})^2 = \Sigma X^2 - \frac{(\Sigma X)^2}{n}$$

Let's do calculations

Step 4

$$s_{YX} = \sqrt{\frac{\Sigma(Y-\hat{Y})^2}{n-2}} = \sqrt{_____}$$

- $s_b = \frac{s_{YX}}{\sqrt{\Sigma(X-\bar{X})^2}}$
- $t = \frac{b-\beta_0}{s_b}$

Critical Value from t-table

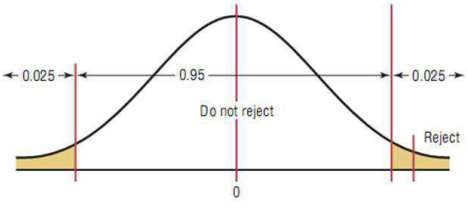
Step 5 : Find the 0.05 column in the top row and 16 in the left-hand column. Where the row and column meet, the appropriate critical value is found.

Table F The *t* Distribution

	Confidence intervals	80%	90%	95%	98%	99%
	One tail, α	0.10	0.05	0.025	0.01	0.005
d.f.	Two tails, α	0.20	0.10	0.05	0.02	0.01
1		3.078	6.314	12.706	31.821	63.657
2		1.886	2.920	4.303	6.965	9.925
3		1.638	2.353	3.182	4.541	5.841
4		1.533	2.132	2.776	3.747	4.604
5		1.476	2.015	2.571	3.365	4.032
6		1.440	1.943	2.447	3.143	3.707
7		1.415	1.895	2.365	2.998	3.499
8		1.397	1.860	2.306	2.896	3.355
9		1.383	1.833	2.262	2.821	3.250
10		1.372	1.812	2.228	2.764	3.169
11		1.363	1.796	2.201	2.718	3.106
12		1.356	1.782	2.179	2.681	3.055
13		1.350	1.771	2.160	2.650	3.012
14		1.345	1.761	2.145	2.624	2.977
15		1.341	1.753	2.131	2.602	2.947
16		1.337	1.746	2.120	2.583	2.921
17		1.333	1.740	2.110	2.567	2.898
18		1.330	1.734	2.101	2.552	2.878
19		1.328	1.729	2.093	2.539	2.861
20		1.325	1.725	2.086	2.528	2.845
21		1.323	1.721	2.080	2.518	2.831
22		1.321	1.717	2.074	2.508	2.819
23		1.319	1.714	2.069	2.500	2.807

Decision Rule

Step 5



Conclusion and Summarizing:

Since the computed value of t=6.89 falls in the critical region, so we reject the null hypothesis and may conclude that there is sufficient reason to say with 95% confidence that height and weight are related.

An other example – Do yourself

In linear regression problem, the following sums were computed from a random sample of size 10.

$\Sigma X = 320, \Sigma Y = 250, \ \Sigma X^2 = 12400, \ \Sigma XY = 9415 \ and \ \Sigma Y^2 = 7230$

Using 5% level of significance, test the hypothesis that the population regression coefficient, β is greater than 0.5.

Hint:

Hint

$H_0 : \beta \quad 0$

$H_1 : \beta > 0$ (Given claim)

T

$est\ formula = t = \frac{b-\beta_0}{s_b}$

- $s_b = \frac{s_{YX}}{\sqrt{\Sigma(X-\bar{X})^2}}$

$b = \frac{n\Sigma XY - (\Sigma X)(\Sigma Y)}{n\Sigma X^2 - (\Sigma X)^2}$

$s_{YX} = \sqrt{\frac{\Sigma(Y-\hat{Y})^2}{n-2}} = \sqrt{\frac{\Sigma Y^2 - a\Sigma Y - b\Sigma XY}{n-2}}$

$a = \bar{Y} - b\bar{X}$

$\Sigma(X-\bar{X})^2 = \Sigma X^2 - \frac{(\Sigma X)^2}{n}$

Summary

- Remember the all hypothesis-testing situations using the traditional method should include the

$$Test\ Value = \frac{(Observed\ value) - (Expected\ value)}{Standard\ error}$$

1. State the null and alternative hypotheses and identify the claim.
2. State an alpha level and find the critical value(s).
3. Compute the test value.
4. Make the decision to reject or not reject the null hypothesis.
5. Summarize the results.

The END

Get Ready …

- Check mic position
- Check resolution size
- Check pen
- Say something….

STA404
Regression & Correlation Analysis
Lesson 9-1

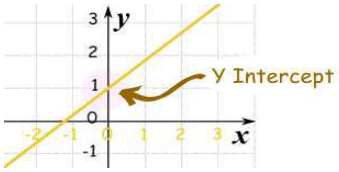
Department of Statistics
Virtual University of Pakistan

Statistical Inference about Regression

Using the t-test for hypothesis testing of
α – the intercept of regression line

What is intercept of regression line?

- The Y intercept of a straight line is simply where the line crosses (cut) the Y axis.

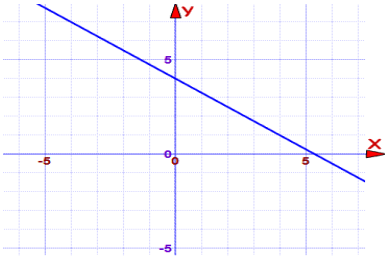


$Y = a + bX$

- In the above diagram the line crosses the Y axis at 1.
- So the Y intercept is equal to 1.

Can you tell?

- What is intercept of the given line?



Difference between Population & Sample Regression

- Population Regression

$$Y = \alpha + \beta X + \varepsilon$$

- Sample Regression

$$Y = a + bX + e$$

Hypothesis Testing – General Procedure

Procedure Table

Solving Hypothesis-Testing Problems (Traditional Method)

Step 1	State the hypotheses and identify the claim.
Step 2	Find the critical value(s) from the appropriate table
Step 3	Compute the test value.
Step 4	Make the decision to reject or not reject the null hypothesis.
Step 5	Summarize the results.

Testing Hypothesis – Example

- For the Given data:
- Test the hypothesis that

$H_0: \alpha = 32$ against $H_1: \alpha \neq 32$

X	Y
65	85
50	74
55	76
65	90
55	85
70	87
65	94
70	98
55	81
70	91
50	76
55	74

Testing Hypothesis – Example

Step 1

Stat the hypothesis.

$H_0: \alpha = 32$

$H_1: \alpha \neq 32$

Step 2

We set the **level of significance (alpha)** $\alpha = 0.01$ **(1%)**

Step 3

The test-statistic (formula) to be used is: $Test\ Value = \frac{(Observed\ value) - (Expected\ value)}{Standard\ error}$

$$t = \frac{a - \alpha_0}{S.E(a)}$$

It follows t-distribution with degree of freedom ***df = n-2***

What we need to calculate?

Do necessary computations from data and solve the formula using these values.

$$t = \frac{a - \alpha_0}{S.E(a)}$$

$$\text{S.E(a)} = s_a = s_{YX}\sqrt{\frac{1}{n} + \frac{\bar{X}^2}{\sqrt{\Sigma(X-\bar{X})^2}}}$$

$$b = \frac{n\Sigma XY - (\Sigma X)(\Sigma Y)}{n\Sigma X^2 - (\Sigma X)^2}$$

$$s_{YX} = \sqrt{\frac{\Sigma(Y-\hat{Y})^2}{n-2}} = \sqrt{\frac{\Sigma Y^2 - a\Sigma Y - b\Sigma XY}{n-2}}$$

$$a = \bar{Y} - b\bar{X}$$

$$\Sigma(X-\bar{X})^2 = \Sigma X^2 - \frac{(\Sigma X)^2}{n}$$

Computation

X	Y	XX	YY	XY
65	85	4225	7225	5525
50	74	2500	5476	3700
55	76	3025	5776	4180
65	90	4225	8100	5850
55	85	3025	7225	4675
70	87	4900	7569	6090
65	94	4225	8836	6110
70	98	4900	9604	6860
55	81	3025	6561	4455
70	91	4900	8281	6370
50	76	2500	5776	3800
55	74	3025	5476	4070
725	1011	44475	85905	61685

Let's do calculations

Step 4

$$b = \frac{n\Sigma XY - (\Sigma X)(\Sigma Y)}{n\Sigma X^2 - (\Sigma X)^2} = _____ = _____ =$$

$$a = \bar{Y} - b\bar{X} = \qquad\qquad =$$

$$\Sigma(Y-\hat{Y})^2 = \Sigma Y^2 - a\Sigma Y - b\Sigma XY$$

$$\Sigma(X-\bar{X})^2 = \Sigma X^2 - \frac{(\Sigma X)^2}{n}$$

Let's do calculations

Step 4

$$s_{YX} = \sqrt{\frac{\Sigma(Y-\hat{Y})^2}{n-2}} = \sqrt{\qquad\qquad}$$

S.E(a) = $s_a = s_{YX}\sqrt{\frac{1}{n} + \frac{\bar{X}^2}{\sqrt{\Sigma(X-\bar{X})^2}}}$

$$t = \frac{a - \alpha_0}{S.E(a)}$$

Critical Value from t-table

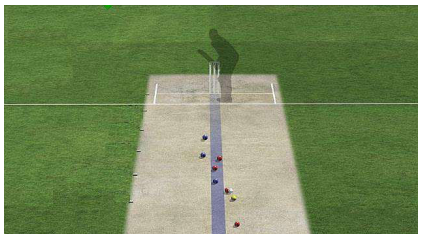
Step 5 : Find the 0.01 column in the top row and 10 in the left-hand column. Where the row and column meet, the appropriate critical value is found.

Table F The *t* Distribution

d.f.	Confidence intervals	80%	90%	95%	98%	99%
	One tail, α	0.10	0.05	0.025	0.01	0.005
	Two tails, α	0.20	0.10	0.05	0.02	0.01
1		3.078	6.314	12.706	31.821	63.657
2		1.886	2.920	4.303	6.965	9.925
3		1.638	2.353	3.182	4.541	5.841
4		1.533	2.132	2.776	3.747	4.604
5		1.476	2.015	2.571	3.365	4.032
6		1.440	1.943	2.447	3.143	3.707
7		1.415	1.895	2.365	2.998	3.499
8		1.397	1.860	2.306	2.896	3.355
9		1.383	1.833	2.262	2.821	3.250
10		1.372	1.812	2.228	2.764	3.169
11		1.363	1.796	2.201	2.718	3.106
12		1.356	1.782	2.179	2.681	3.055
13		1.350	1.771	2.160	2.650	3.012
14		1.345	1.761	2.145	2.624	2.977
15		1.341	1.753	2.131	2.602	2.947
16		1.337	1.746	2.120	2.583	2.921
17		1.333	1.740	2.110	2.567	2.898
18		1.330	1.734	2.101	2.552	2.878
19		1.328	1.729	2.093	2.539	2.861
20		1.325	1.725	2.086	2.528	2.845
21		1.323	1.721	2.080	2.518	2.831
22		1.321	1.717	2.074	2.508	2.819
23		1.319	1.714	2.069	2.500	2.807

Decision Rule

Step 5



Conclusion and Summarizing:

Since the computed value of **t = -0.169** does NOT fall in the critical region, so we DO NOT reject the null hypothesis.

An other example – Do yourself

In linear regression problem, the following sums were computed from a random sample of size 10.

$$\Sigma X = 320,\ \Sigma Y = 250,\ \Sigma X^2 = 12400,\ \Sigma XY = 9415\ and\ \Sigma Y^2 = 7230$$

$$\hat{Y} = 4.04 + 0.655X$$

$$\Sigma(Y-\hat{Y})^2 = 53.175$$

$$\Sigma(X-\bar{X})^2 = 2160$$

Using **5% level** of significance, test the hypothesis that the intercept of regression line, α is zero.

Hint:

Hint

$H_0:\ \alpha = 0$ (Given claim)

$H_1:\ \alpha \neq 0$

T

$$\hat{Y} = 4.04 + 0.655X$$

Test formula = $t = \frac{a-\alpha_0}{S.E(a)}$

$$b = \frac{n\sum XY - (\sum X)(\sum Y)}{n\sum X^2 - (\sum X)^2}$$

S.E(a) = $s_a = s_{YX}\sqrt{\frac{1}{n} + \frac{\bar{X}^2}{\sqrt{\Sigma(X-\bar{X})^2}}}$

$$s_{YX} = \sqrt{\frac{\Sigma(Y-\hat{Y})^2}{n-2}} = \sqrt{\frac{\sum Y^2 - a\sum Y - b\sum XY}{n-2}}$$

$$\Sigma(X-\bar{X})^2 = \Sigma X^2 - \frac{(\Sigma X)^2}{n}$$

Summary

- Remember the all hypothesis-testing situations using the traditional method should include the

$$Test\ Value = \frac{(Observed\ value) - (Expected\ value)}{Standard\ error}$$

1. State the null and alternative hypotheses and identify the claim.
2. State an alpha level and find the critical value(s).
3. Compute the test value.
4. Make the decision to reject or not reject the null hypothesis.
5. Summarize the results.

The END

STA404
Regression & Correlation Analysis
Lesson 10-1

Department of Statistics
Virtual University of Pakistan

Statistical Inference about Simple Correlation

Testing hypothesis that the population correlation coefficient ρ is equal to some specified value other than zero

Hypothesis Testing – General Procedure

Procedure Table

Solving Hypothesis-Testing Problems (Traditional Method)

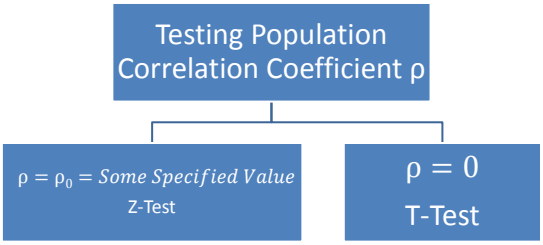
Step 1	State the hypotheses and identify the claim.
Step 2	Find the critical value(s) from the appropriate table
Step 3	Compute the test value.
Step 4	Make the decision to reject or not reject the null hypothesis.
Step 5	Summarize the results.

Notations:

- Sample Correlation Coefficient
- English letter "r"

- Population Correlation Coefficient
- Greek small letter "ρ" (rho)

Two Cases



Some Main Points

- **Problem:** Sampling distribution of "r" is
  - Neither normal
  - Nor becomes approximately normal for large sample size
- **Solution:** Do Transformation:

  Old variable → new variable

"r" ---------- "Z_f"

$$"r".....Z_f = 1.1513 \log \frac{1+r}{1-r}$$

Testing formula

$$Test\ Value = \frac{(Observed\ value) - (Expected\ value)}{Standard\ error}$$

$$Z = \frac{Z_f - \mu_z}{1/\sqrt{n-3}}$$

Where,

$$Z_f = 1.1513 \log \frac{1+r}{1-r}$$

$$\mu_z = 1.1513 \log \frac{1+\rho}{1-\rho}$$

Testing Hypothesis – Example

- A random sample of 28 pairs from a bivariate normal population showed a correlation coefficient of 0.7.
- Is this value consistent with the assumption that the correlation coefficient in the population is 0.5?

Testing Hypothesis – Example

Step 1

Stat the hypothesis.

H_0 : ρ = 0.5 (Given claim)

H_1 :

Step 2

We set the **level of significance (alpha)** $\alpha = 0.05$ **(5%)**

Step 3

The test-statistic (formula) to be used is: $Test\ Value = \frac{(Observed\ value) - (Expected\ value)}{Standard\ error}$

$$Z = \frac{Z_f - \mu_z}{1/\sqrt{n-3}}$$

– *It is z-test.*

Calculations

$$Z = \frac{Z_f - \mu_z}{1/\sqrt{n-3}}$$

Given : $r = 0.7,\ \rho = 0.5$ and $n = 28$

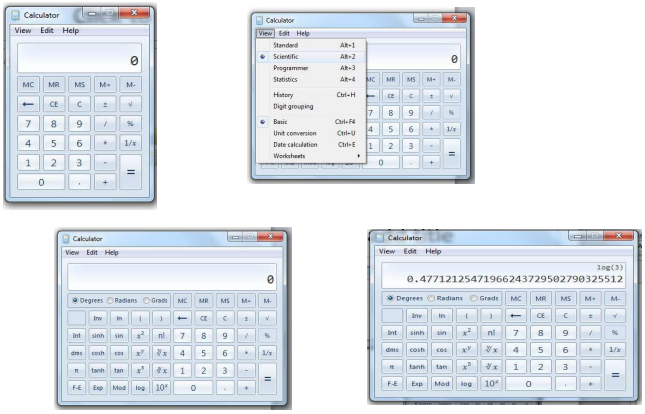
$$Z_f = 1.1513\ \log\frac{1+r}{1-r} = 1.1513\ \log\frac{1+0.7}{1-0.7} = 1.1513\ \log\frac{1.7}{0.3}$$

$$Z_f = 1.1513\ \log(5.66) = 1.1513\ (0.7528) = 0.87$$

$$\mu_z = 1.1513\ \log\frac{1+\rho}{1-\rho} =$$

Calculating log Value

Windows calculator

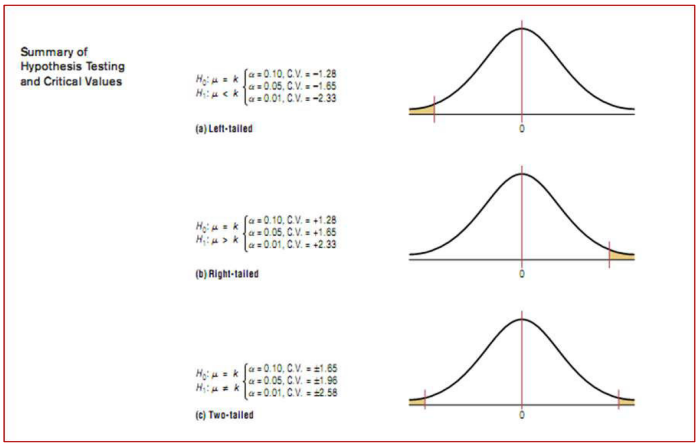


Calculations

Now we have: $r = 0.7,\ n = 28,\ Z_f = 0.87,\ \mu_f = 0.55$

$$Z = \frac{Z_f - \mu_z}{1/\sqrt{n-3}}$$

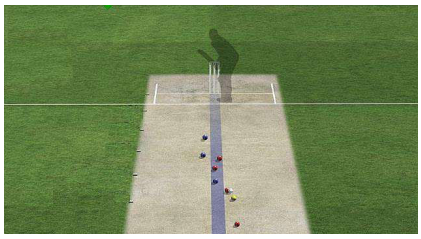
Frequently used Z Critical Values



Decision Rule

Compare Calculated value with Table Value

Step 5



Conclusion and Summarizing:

Since the computed value of **z = 1.60** does NOT fall in the critical region, so we do not reject our null hypothesis and may conclude that the given claim is correct.

Summary

- When **ρ ≠ 0**, then distribution of "r" is
  - Neither normal
  - Nor becomes normal on large sample
- We transform "r" into a new variable which is distributed normally. Hence we can use to for testing.

$$Z = \frac{Z_f - \mu_z}{1/\sqrt{n-3}}$$

- Its z-test.
- Rest of the testing procedure is same.

The END

Get Ready …

- Check mic position
- Check resolution size
- Check pen
- Say something….

STA404
Regression & Correlation Analysis
Lesson 10-2

Department of Statistics
Virtual University of Pakistan

Statistical Inference about Simple Correlation

Testing hypothesis that the population correlation coefficient ρ is equal to zero

Hypothesis Testing – General Procedure

Procedure Table

Solving Hypothesis-Testing Problems (Traditional Method)

Step 1	State the hypotheses and identify the claim.
Step 2	Find the critical value(s) from the appropriate table
Step 3	Compute the test value.
Step 4	Make the decision to reject or not reject the null hypothesis.
Step 5	Summarize the results.

Notations:

- Correlation Coefficient from SAMPLE data
- English letter "r"

- Correlation Coefficient from POPULATION data
- Greek small letter "ρ" (rho)

Two Cases

Testing Population Correlation Coefficient ρ

- $\rho = \rho_0 = Some\ Specified\ Value$ — Z-Test
- $\rho = 0$ — T-Test

Some Main Points

- Sampling distribution is NOT symmetry BUT it is interesting to note that when ρ = 0, then its symmetrical.
- Above make it possible to use the t-test.
- T-test can be applied for any sample size
- What does it means Ho: ρ = 0
  - There is NO linear correlation between two variables

Testing Hypothesis – Example

- A random sample of 20 pairs of observations gives a coefficient of 0.45.
- Test the hypothesis at the 0.05 level of significance that the correlation coefficient in the population is zero.

Testing Hypothesis – Example

Step 1

Stat the hypothesis.

$H_0: \rho = 0$ (There is NO linear correlation between two variables)

$H_1: \rho \neq 0$

Step 2

We set the **level of significance (alpha)** $\alpha = 0.05$ **(5%)**

Step 3

The test-statistic (formula) to be used is: $Test\ Value = \frac{(Observed\ value) - (Expected\ value)}{Standard\ error}$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

It follows t-distribution with degree of freedom ***df = n-2***

Calculations

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

$$t = \frac{0.45\sqrt{20-2}}{\sqrt{1-(0.45)^2}}$$

$$t = \frac{1.91}{0.89} = 2.14$$

t = t-Calcuated Value = 2.14

Critical Value from t-table

Step 5 : Find the 0.05 column in the top row and 18 in the left-hand column. Where the row and column meet, the appropriate critical value is found.

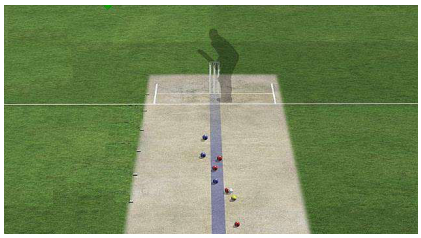
Table F The *t* Distribution

	Confidence intervals	80%	90%	95%	98%	99%
	One tail, α	0.10	0.05	0.025	0.01	0.005
d.f.	Two tails, α	0.20	0.10	0.05	0.02	0.01
1		3.078	6.314	12.706	31.821	63.657
2		1.886	2.920	4.303	6.965	9.925
3		1.638	2.353	3.182	4.541	5.841
4		1.533	2.132	2.776	3.747	4.604
5		1.476	2.015	2.571	3.365	4.032
6		1.440	1.943	2.447	3.143	3.707
7		1.415	1.895	2.365	2.998	3.499
8		1.397	1.860	2.306	2.896	3.355
9		1.383	1.833	2.262	2.821	3.250
10		1.372	1.812	2.228	2.764	3.169
11		1.363	1.796	2.201	2.718	3.106
12		1.356	1.782	2.179	2.681	3.055
13		1.350	1.771	2.160	2.650	3.012
14		1.345	1.761	2.145	2.624	2.977
15		1.341	1.753	2.131	2.602	2.947
16		1.337	1.746	2.120	2.583	2.921
17		1.333	1.740	2.110	2.567	2.898
18		1.330	1.734	2.101	2.552	2.878
19		1.328	1.729	2.093	2.539	2.861
20		1.325	1.725	2.086	2.528	2.845
21		1.323	1.721	2.080	2.518	2.831
22		1.321	1.717	2.074	2.508	2.819
23		1.319	1.714	2.069	2.500	2.807

Decision Rule

Compare Calculated value with Table Value

Step 5



Conclusion and Summarizing:

Since the computed value of **t = 2.14** falls in the critical region, so we reject our null hypothesis and may conclude that the correlation coefficient in the population differ from zero.

It is similar to test the ….

- Ho: ρ = 0
  - There is NO linear correlation between two variables.
  - The two variables are INDEPENDET
- Ho: β = 0
  - There is NO linear correlation between two variables.
  - The two variables are INDEPENDET

Summary

- When H_0 : **ρ = 0**, then distribution of "r" is symmetry. Therefor we can use t-test
- The formula is: $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$
- Testing **H_0 : ρ = 0** is equivalent to test **H_0 : β = 0**
- T-test can be applied to any sample size.

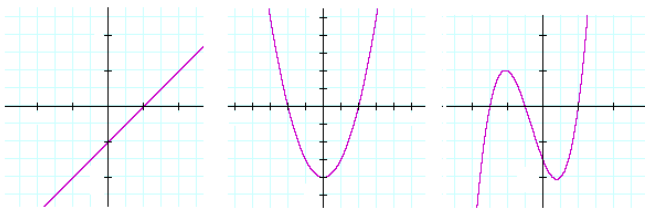
The END

Regression & Correlation Analysis

How to fit a Second Degree Parabola line?

Setting up Normal Equations

Shape of equations/data …



Normal Equations?

- Normal equations are equations obtained by setting equal to zero the partial derivatives of the sum of squared errors (**least squares**)
- This is a technique for computing coefficients for equations
- It finds the regression coefficients ***analytically***

How to make Normal Equations …

$$Y = a + bX \qquad Y = a + bX + cX^2$$

The normal Equations are …..

$$Y = a + bX \text{ ----> } \Sigma Y = na + b\Sigma X$$

$$XY = aX + bX^2 \text{ ---> } \Sigma XY = a\Sigma X + b\Sigma X^2$$

How to make Normal Equations …

$$Y = a + bX + cX^2$$

The normal Equations are …..

$1\left(Y = a + bX + cX^2\right)$ • $\Sigma Y =$

$X\left(Y = a + bX + cX^2\right)$ • $\Sigma XY =$

$X^2\left(Y = a + bX + cX^2\right)$ • $\Sigma X^2 Y =$

Test your Knowledge …

Can you find normal equation for the following equation ?

$$Y = a + bX + cX^2 + dX^3$$

- $\Sigma Y =$
- $\Sigma XY =$
- $\Sigma X^2 Y =$
- $\Sigma X^3 Y =$

- The END -

Regression & Correlation Analysis

How to fit a Second Degree Parabola line?

Procedure and step by step calculation with example

How to fit a Second Degree Parabola?

Example:

Fit a **second degree parabola** to the following data, taking X as independent variable.

X	Y
0	1
1	1.8
2	1.3
3	2.5
4	6.3

Equation of Second Degree …

$$Y = a + bX + cX^2$$

The normal Equations are …..

- $\Sigma Y = na + b\Sigma X + c\Sigma X^2$
- $\Sigma XY = a\Sigma X + b\Sigma X^2 + c\Sigma X^3$
- $\Sigma X^2 Y = a\Sigma X^2 + b\Sigma X^3 + c\Sigma X^4$

What we need…?

Required Calculation

X	Y
0	1.0
1	1.8
2	1.3
3	2.5
4	6.3
Total	

Substituting the values in the normal Equations…..

$$\Sigma Y = na + b\Sigma X + c\Sigma X^2$$
$$\Sigma XY = a\Sigma X + b\Sigma X^2 + c\Sigma X^3$$
$$\Sigma X^2 Y = a\Sigma X^2 + b\Sigma X^3 + c\Sigma X^4$$

Solve these equation to eliminate "a"

$$12.9 = 5a + 10b + 30c$$
$$37.1 = 10a + 30b + 100c$$
$$130.3 = 30a + 100b + 354c$$

Now, Solve these equation to eliminate "b"

$$11.3 = 10b + 40c$$
$$19 = 10b + 54c$$

Now, put value of "b" and "c" in Eq…1

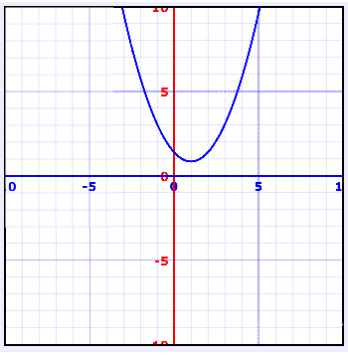
$$12.9 = 5a + 10b + 30c$$

So by solving three normal equations simultaneously, we are able to find the values

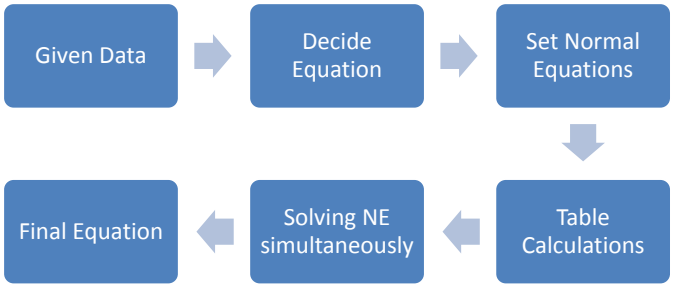
$$a =$$
$$b =$$
$$c =$$

$$Y = a + bX + cX^2$$

Just for fun … Graph of this equation



Recap



- The END -

Test 1…2…3

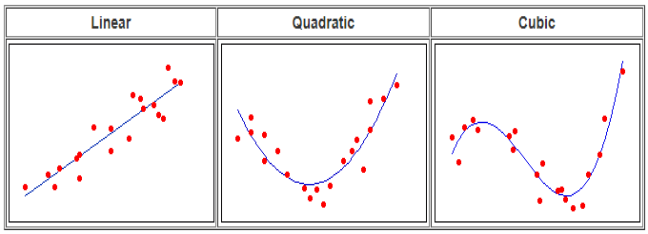
- Mic/Audio test
- Screen Area test
- Pen test

Regression & Correlation Analysis

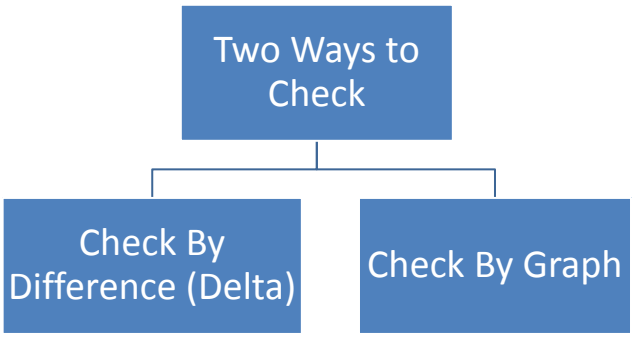
Deciding the Type of Curve to fit

Which curve is suitable in which situation

Lines/Curves come in different shapes ….



Choosing the right line…



Example #1

Question:

Decide the **suitable** curve for the following data.

X	Y
1	12
2	15
3	18
4	21

Lets do it ….

Solution:

Let, first we take its first difference as below and see whether it is constant or not?

X	Y	First Difference
1	12	-
2	15	15 – 12 =
3	18	18 – 15 =
4	21	21 – 18 =

Delta Y

Criteria for suitable curve

Some more rules:

- If the FIRST DIFFERENCES of Y are ***approximately*** **constant**, use a straight line
- If the SECOND DIFFERENCES of Y are ***approximately*** **constant**, use a second degree order equation (parabola)
- If the THIRD DIFFERENCES of Y are ***approximately*** **constant**, use a third degree order equation

Criteria for suitable curve

Here are few rules:

- If the FIRST DIFFERENCES of **log Y** are *approximately* constant, use a exponential curve
- If the FIRST DIFFERENCES of **log Y** and **log X** are *approximately* constant, use a Geometric curve
- If the FIRST DIFFERENCES of **reciprocals of Y** are *approximately* constant, use a third degree parabola

Lets do it ….

SOLUTION:

– we take its first difference

– and see whether it is constant or not?

– If not then go for second difference

X	Y	First Difference	
1	10	--	--
2	14	14 – 10 =	
3	21	21 – 14 =	
4	31	31 – 21 =	
5	44	44 – 31 =	

Remember

- Straight Line:
  $Y = a + bX$
- Second Degree line (Also called Parabola)
  $Y = a + bX + cX^2$
- Third Degree Line
  $Y = a + bX + cX^2 + dX^3$

- The END -

Example #2 ….

Question:

Decide the **suitable** curve for the following data.

X	Y
1	10
2	14
3	21
4	31
5	44

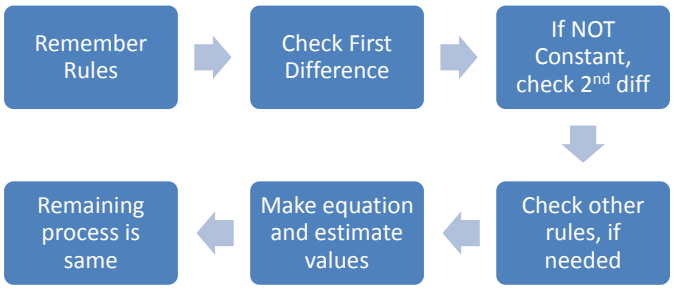
Example #3 ….

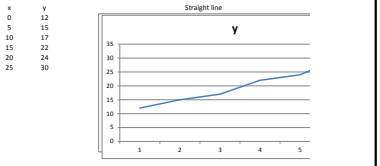
Question:

Suggest a suitable curve for the given data?

X	Y
0	10
1	17
2	28
3	43
4	62

Recap







Test 1…2…3

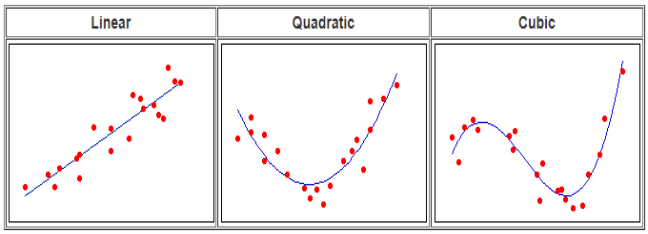
- Mic/Audio test
- Screen Area test
- Pen test

Regression & Correlation Analysis

Deciding the Type of Curve to fit

Use graph to get the idea of suitable equation

Lines/Curves come in different shapes ….



Criteria for suitable curve

Here are few guidelines:

- If the graph of the values of **X and Y** gives a straight line, use a straight line .
- If the graph of the values of **X and Y** gives a curve with only one bend, use a second degree parabola.
- If the graph of the values of **X and Y** gives a curve of the shape of reverse "S", use a third degree parabola.

Criteria for suitable curve

Few more guidelines:

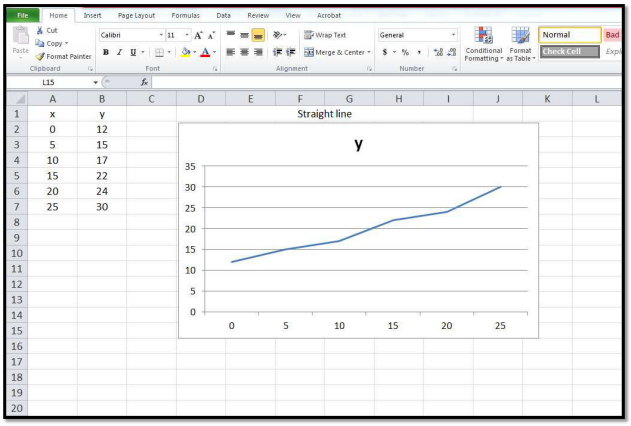
- If the graph of the values of **X and log Y** gives a straight line, use an exponential curve .
- If the graph of the values of **log X and log Y** gives a straight line, use a geometric or logarithmic curve .
- If the graph of the values of **X and 1/Y** gives a straight line, use a hyperbola curve.

Lets explore few … Example #1

Question:
By plotting, which type of line/curve is appropriate for the data?

x	y
0	12
5	15
10	17
15	22
20	24
25	30

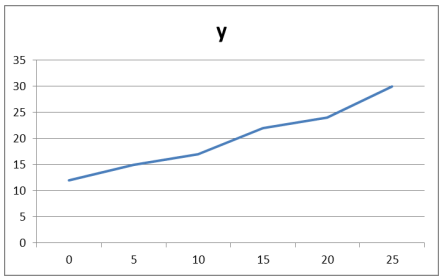
Lets Go To Excel ….



Straight line …

Solution:

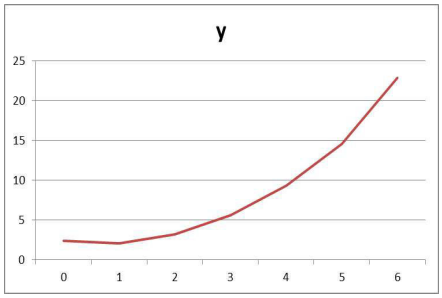
- If the graph of the values of **X and Y** gives a straight line, use a straight line .



Second Degree Parabola …

SOLUTION:

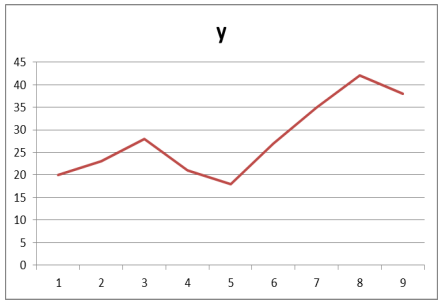
- If the graph of the values of **X and Y** gives a curve with only one bend, use a second degree parabola.



Third Degree Parabola…

SOLUTION:

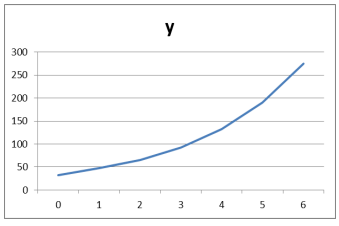
- If the graph of the values of **X and Y** gives a curve of the shape of reverse "S", use a third degree parabola.

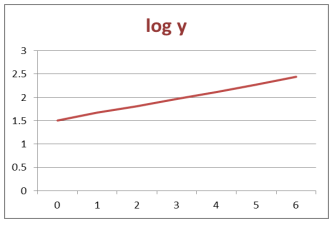


Exponential Curve …

SOLUTION:

- If the graph of the values of **X and log Y** gives a straight line, use an exponential curve .





Example #2 ….

Question:

Decide the **suitable** curve for the following data.

X	Y
0	2.4
1	2.1
2	3.2
3	5.6
4	9.3
5	14.6
6	22.9

Example #3 ….

Question:

Decide the **suitable** curve for the following data.

X	Y
1	20
2	23
3	28
4	21
5	18
6	27
7	35
8	42
9	38

Example #4 ….

Question:

Suggest a suitable curve for the given data?

x	y
0	32
1	47
2	65
3	92
4	132
5	190
6	275

Remember

- Straight Line:
  $$Y = a + bX$$
- Second Degree line (Also called Parabola)
  $$Y = a + bX + cX^2$$
- Third Degree Line
  $$Y = a + bX + cX^2 + dX^3$$

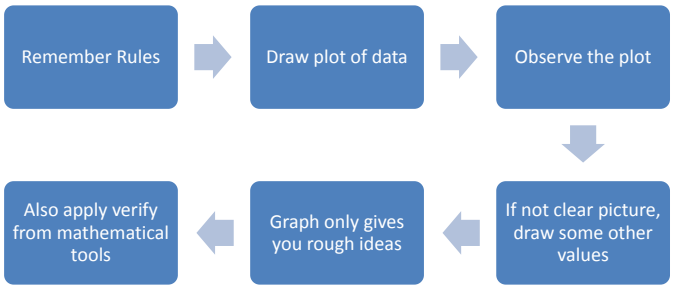
Remember

- Exponential form:
  $Y = ab^X \quad >>> \quad \log Y = \log a + X \log b$
- Other exponential form

  $Y = ae^{bX} \quad >>> \quad \log Y = \log a + (b \log e) X$
- Reciprocal form

  $\frac{1}{Y} = a + bX \quad >>> \quad Y' = a + bX$

Recap



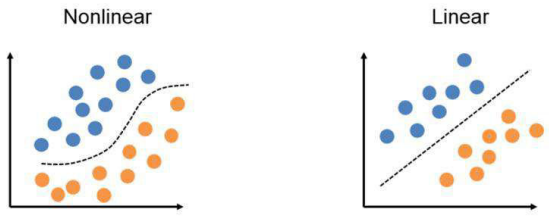
- The END -

Regression & Correlation Analysis

Converting non-linear into linear form

Some useful transformations to convert non-linear equations into linear form

Non-linear VS Linear



#1

Non-linear form	$Y = aX^b$
Transformation	Taking log on both sides,
Linear form	$Y' = A + bX'$

#2

Non-linear form	$Y = ab^X$
Transformation	Taking log on both sides,
Linear form	$Y' = A + BX$

#3

Non-linear form	$\frac{1}{Y} = a + bX$
Transformation	Tranform $\frac{1}{Y}$ only...
Linear form	$Y' = a + bX$

#4

Non-linear form	$Y = \frac{1}{a + bX}$
Transformation	Take Reciprocal / Tranform Y only...
Linear form	$Y' = a + bX$

#5

Non-linear form	$\frac{1}{Y} = a + \frac{b}{1+X}$
Transformation	Change $\frac{1}{Y}$ and $\frac{1}{1+X}$
Linear form	$Y' = a + bX'$

#6

Non-linear form	$Y = a + b\sqrt{X}$
Transformation	Let $\sqrt{X}$ =
Linear form	$Y = a + bX'$

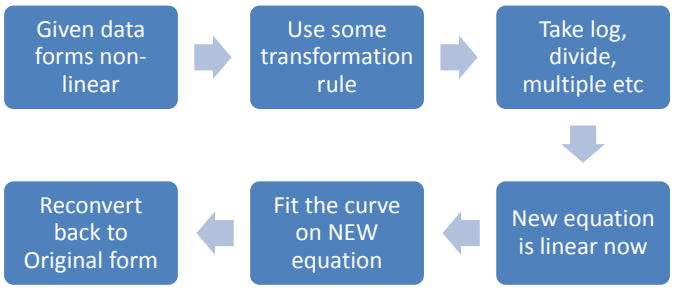
#7

Non-linear form	$Y = aX^2 + bX$
Transformation	Divide all by X
Linear form	$Y' = aX + b$

Summary

Non-linear form	Let/Suppose/Transformation	Linear form
$Y = aX^b$	$Y' = \log Y, A = \log a, X' = \log X$	$Y' = A + bX'$
$Y = ab^X$	$Y' = \log Y, A = \log a, B = \log b$	$Y' = A + BX$
$Y = \frac{1}{a + bX}$	$Y' = \frac{1}{Y}$	$Y' = a + bX'$
$\frac{1}{Y} = a + bX$	$Y' = \frac{1}{Y}$	$Y' = a + bX'$
$\frac{1}{Y} = a + \frac{b}{1+X}$	$Y' = \frac{1}{Y}, X' = \frac{1}{1+X}$	$Y' = a + bX'$
$Y = a + b\sqrt{X}$	$X' = \sqrt{X}$	$Y' = a + bX'$
$Y = aX^2 + bX$	$Y' = \frac{Y}{X}$	$Y' = aX + b$

Recap



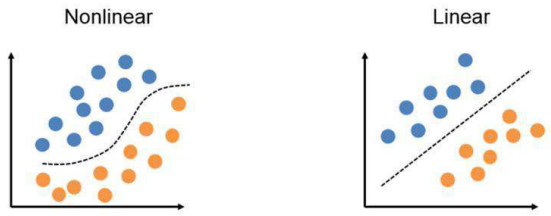
- The END -

Regression & Correlation Analysis

Converting non-linear into linear form

Some useful transformations to convert non-linear equations into linear form

Non-linear VS Linear



#1

Non-linear form	$Y = aX^b$
Transformation	Taking log on both sides,
Linear form	$Y' = A + bX'$

#2

Non-linear form	$Y = ab^X$
Transformation	Taking log on both sides,
Linear form	$Y' = A + BX$

#3

Non-linear form	$\frac{1}{Y} = a + bX$
Transformation	Tranform $\frac{1}{Y}$ only...
Linear form	$Y' = a + bX$

#4

Non-linear form	$Y = \frac{1}{a + bX}$
Transformation	Take Reciprocal / Tranform Y only...
Linear form	$Y' = a + bX$

#5

Non-linear form	$\frac{1}{Y} = a + \frac{b}{1 + X}$
Transformation	Change $\frac{1}{Y}$ and $\frac{1}{1+X}$
Linear form	$Y' = a + bX'$

#6

Non-linear form	$Y = a + b\sqrt{X}$
Transformation	Let $\sqrt{X}$ =
Linear form	$Y = a + bX'$

#7

Non-linear form	$Y = aX^2 + bX$
Transformation	Divide all by X
Linear form	$Y' = aX + b$

Summary

Non-linear form	Let/Suppose/Transformation	Linear form
$Y = aX^b$	$Y' = \log Y, A = \log a, X' = \log X$	$Y' = A + bX'$
$Y = ab^X$	$Y' = \log Y, A = \log a, B = \log b$	$Y' = A + BX$
$Y = \frac{1}{a + bX}$	$Y' = \frac{1}{Y}$	$Y' = a + bX'$
$\frac{1}{Y} = a + bX$	$Y' = \frac{1}{Y}$	$Y' = a + bX'$
$\frac{1}{Y} = a + \frac{b}{1+X}$	$Y' = \frac{1}{Y}, X' = \frac{1}{1+X}$	$Y' = a + bX'$
$Y = a + b\sqrt{X}$	$X' = \sqrt{X}$	$Y' = a + bX'$
$Y = aX^2 + bX$	$Y' = \frac{Y}{X}$	$Y' = aX + b$

- The END -

Recap

